

Understanding Urban India

Population density, housing, transport, air quality and energy in India's cities, read as one story.

5 CHAPTERS	36 INDICATORS	8 WITH STATE DATA	1951–2036 PERIOD COVERED
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Generated 21 Aug 2026 · assembled from 31 sources

EXECUTIVE SUMMARY

This report assembles 36 indicators across 5 chapters, drawn from 31 sources and covering 1951–2036.

Of those indicators, 8 carry state-level detail and can be mapped across India's states. The remaining 28 are published at national level only by their source, and are shown here as national series without a sub-national breakdown.

Coverage limits. This report contains no district- or city-level figures. Comparable district data is not currently integrated into India Insight Lab, and no value here has been estimated, interpolated or carried down from a higher geography to fill that gap. Where a number is absent, it is absent.

KEY FINDINGS

Each finding below is computed directly from the state values in this report — the highest and lowest recorded state for that indicator. None are authored interpretations.

1 **Slum population share by state**

Ranges from 24.0% in Andhra Pradesh to 4.9% in Kerala — a spread of 19.1% across the 15 states reported.

Census of India 2011, Primary Census Abstract for Slum

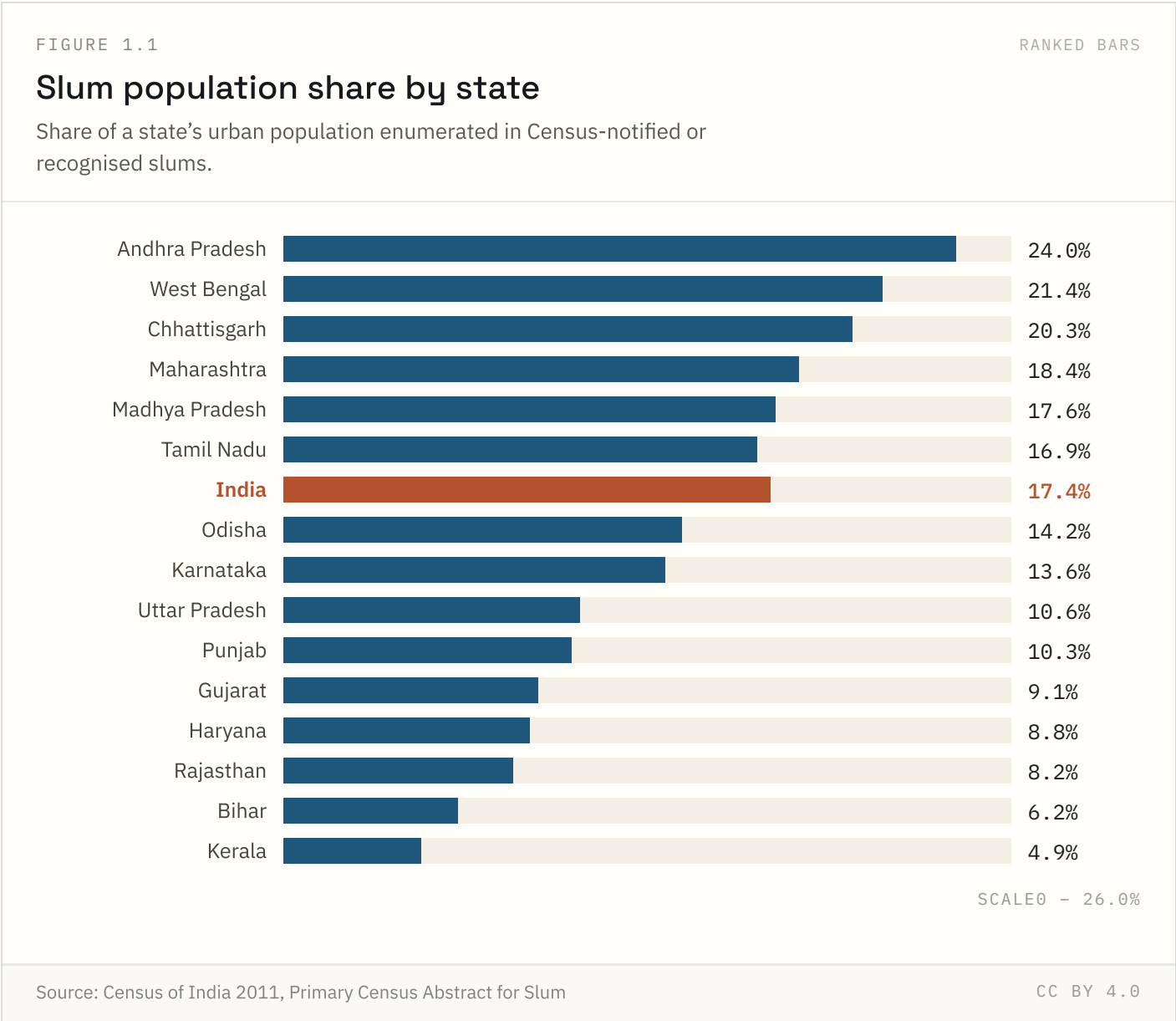
2 **Forest cover by state**

Ranges from 85.4% in Mizoram to 3.6% in Haryana — a spread of 81.8% across the 8 states reported.

India State of Forest Report, Forest Survey of India

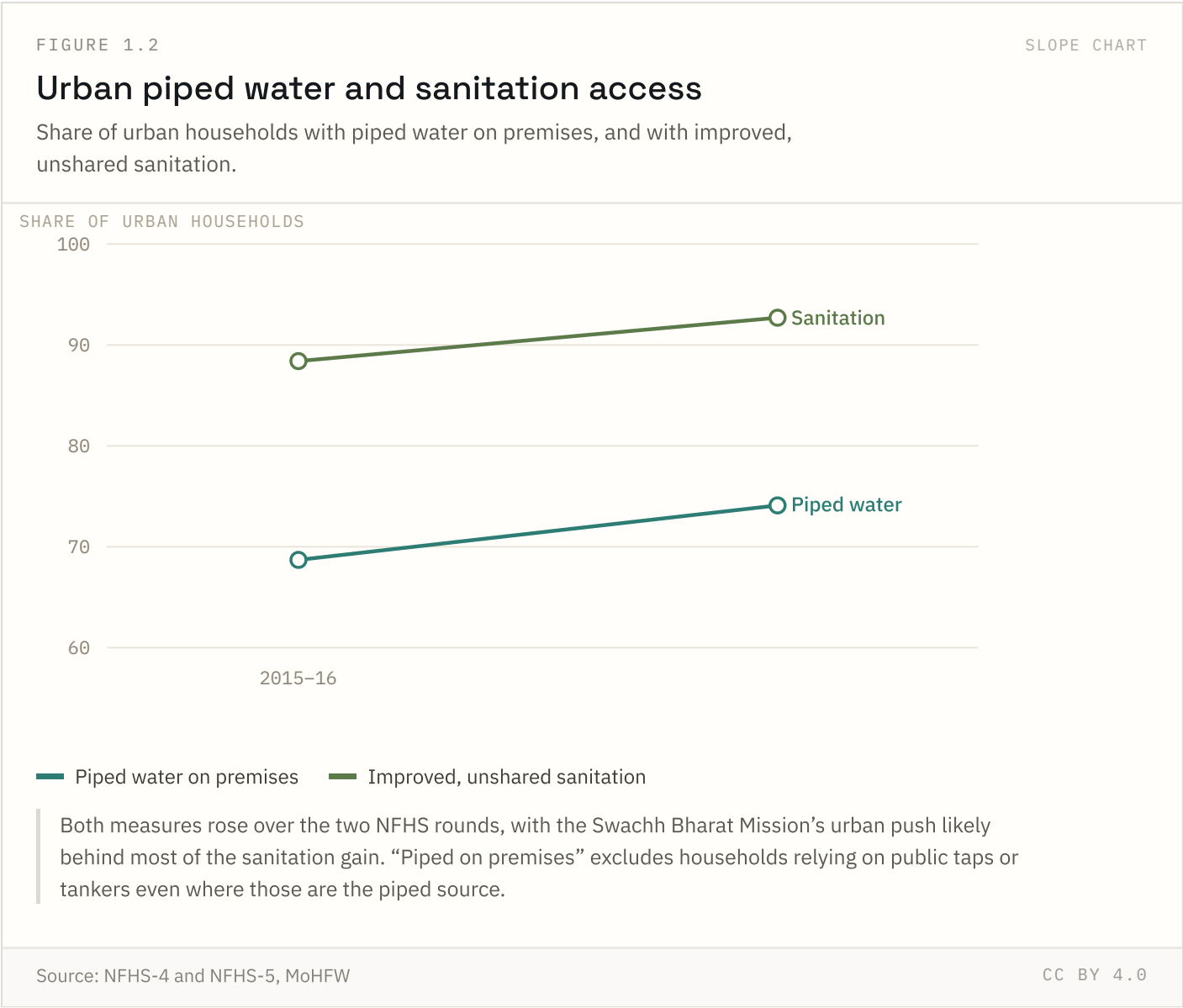
Urban development & housing

1 of 7 indicators in this chapter carry state-level detail; the rest are published at national level only.



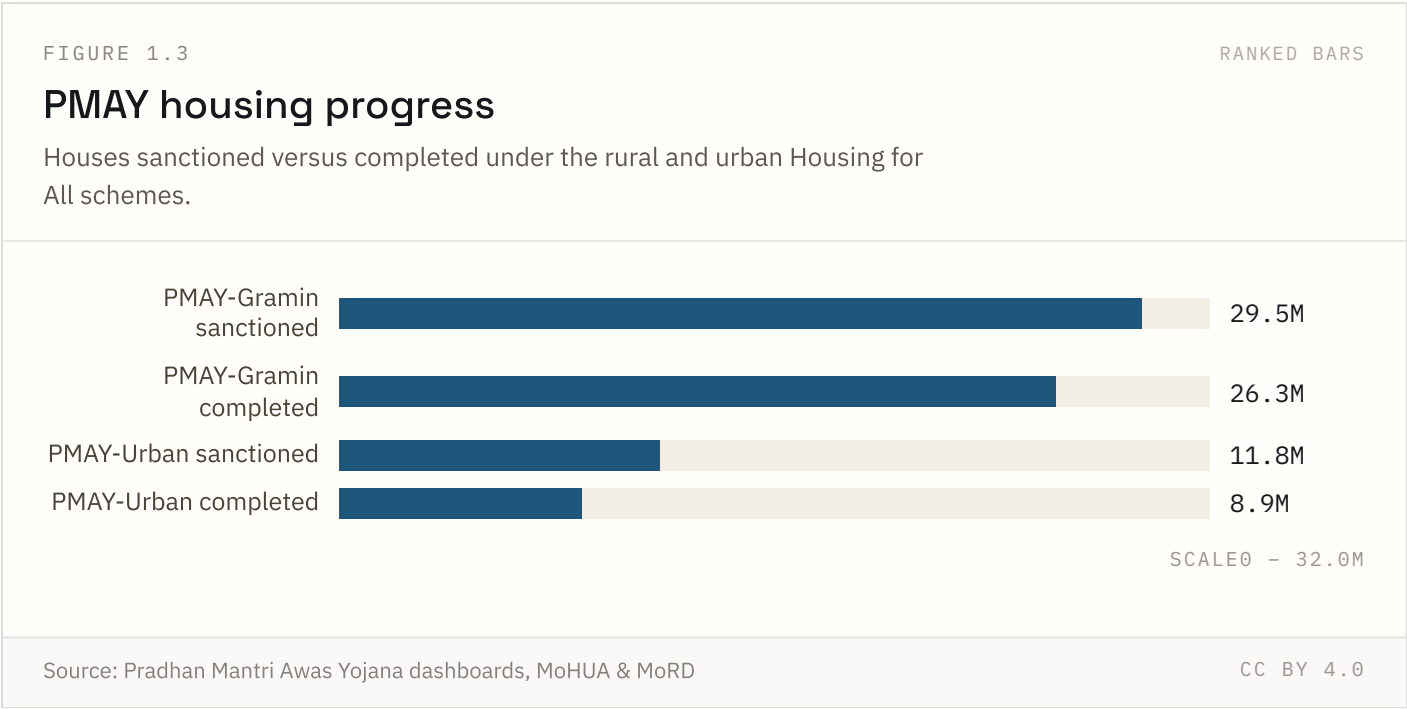
This is the last Census-based slum count available — no Census has been held since 2011, so growth in informal settlements over the past decade is not captured here. Definitions vary by state (notified, recognised and identified slums), which limits strict comparability.

Highest: **Andhra Pradesh** at 24.0%. Lowest: **Kerala** at 4.9%.



Both series improved between NFHS rounds. “Piped on premises” is stricter than “access to an improved water source” — it excludes households that rely on public taps, tankers or neighbours’ connections even if the underlying source is piped.

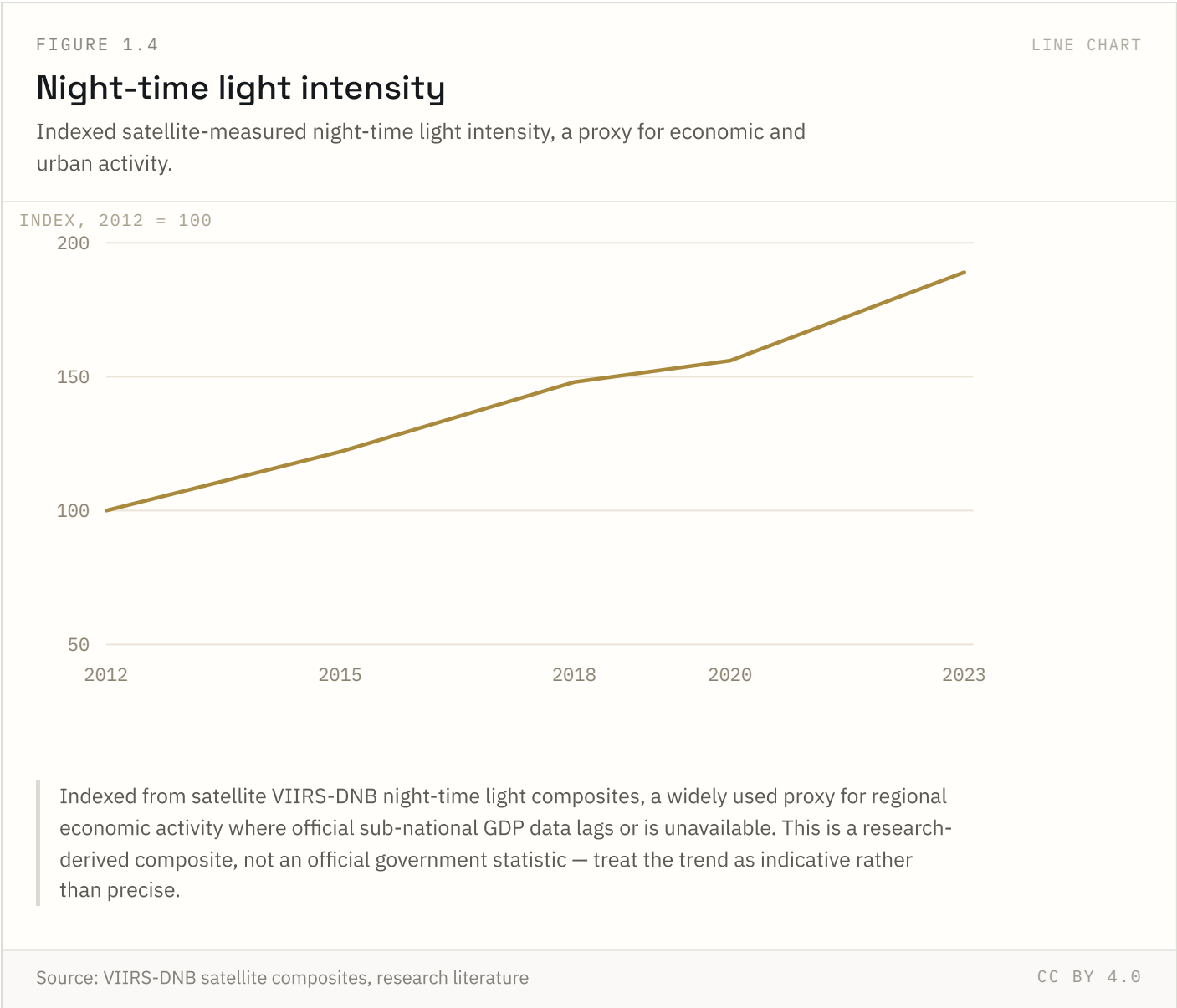
FIGURE 1.3 NATIONAL ONLY



The rural scheme (PMAY-Gramin) is closer to its sanctioned target than the urban scheme (PMAY-Urban), where land acquisition and construction timelines run longer. “Completed” means handed over to the beneficiary, not merely under construction.

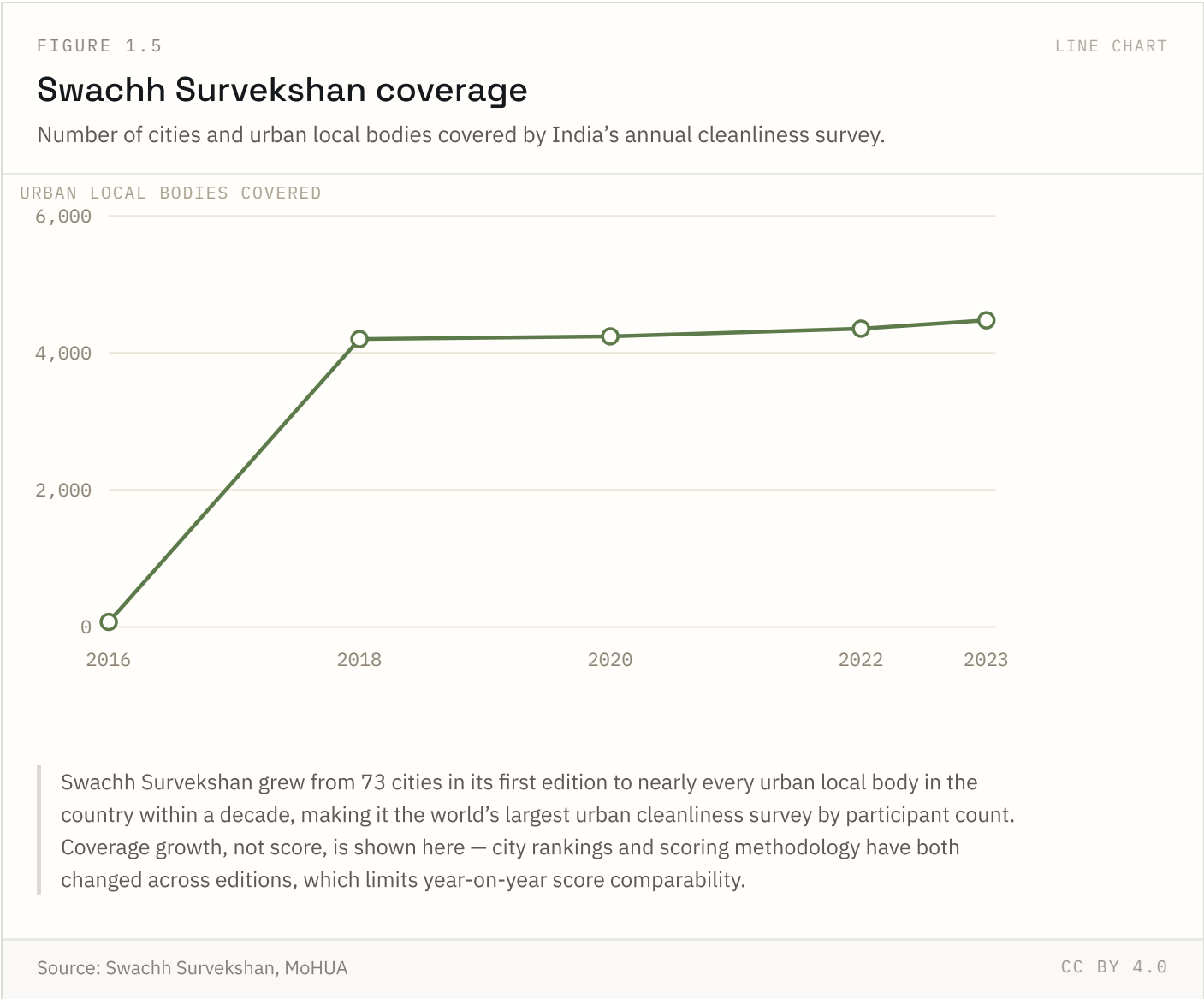
Source: Pradhan Mantri Awas Yojana dashboards, MoHUA & MoRD Coverage: Cumulative to 2025 Updated: 18 Aug 2026

FIGURE 1.4 NATIONAL ONLY



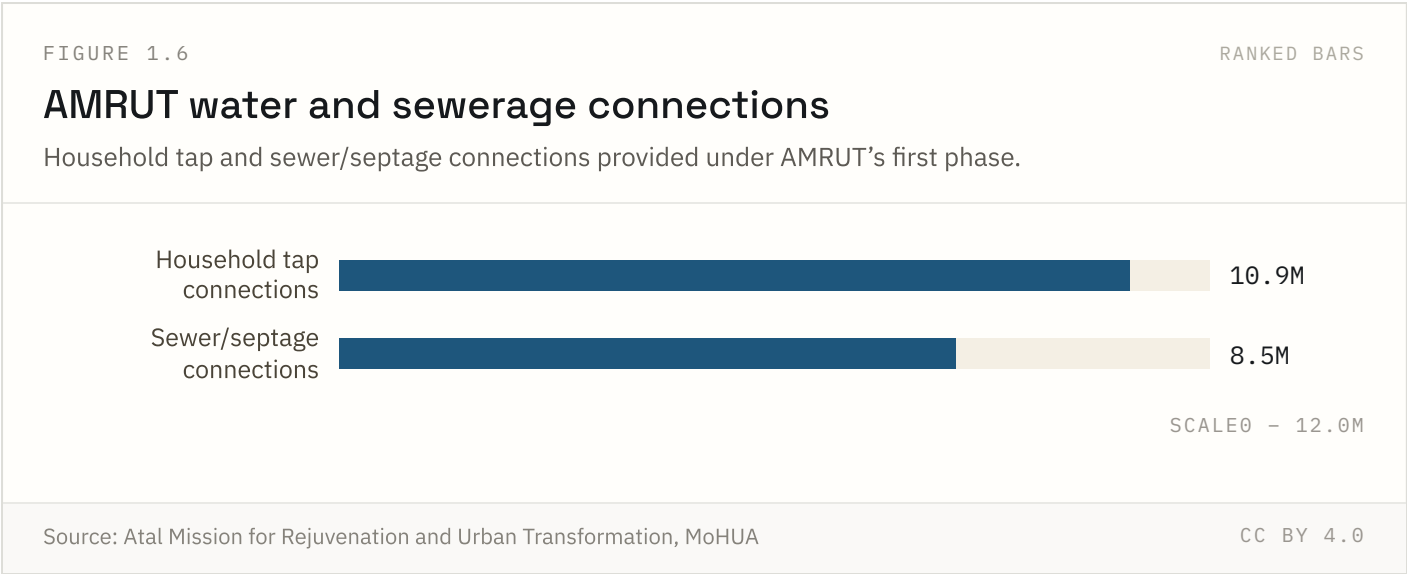
Night-time lights are used by researchers as a fast, sub-national proxy for economic growth where official data is thin or delayed. It captures electrified activity, not output directly, so it can overstate growth in areas that simply gained grid access rather than economic output.

FIGURE 1.5 NATIONAL ONLY



The survey has become close to universal across urban India, which is itself a policy achievement independent of any city’s score. Scoring methodology and weightings have changed across editions, so a city’s rank in different years is not a clean apples-to-apples comparison.

FIGURE 1.6 NATIONAL ONLY

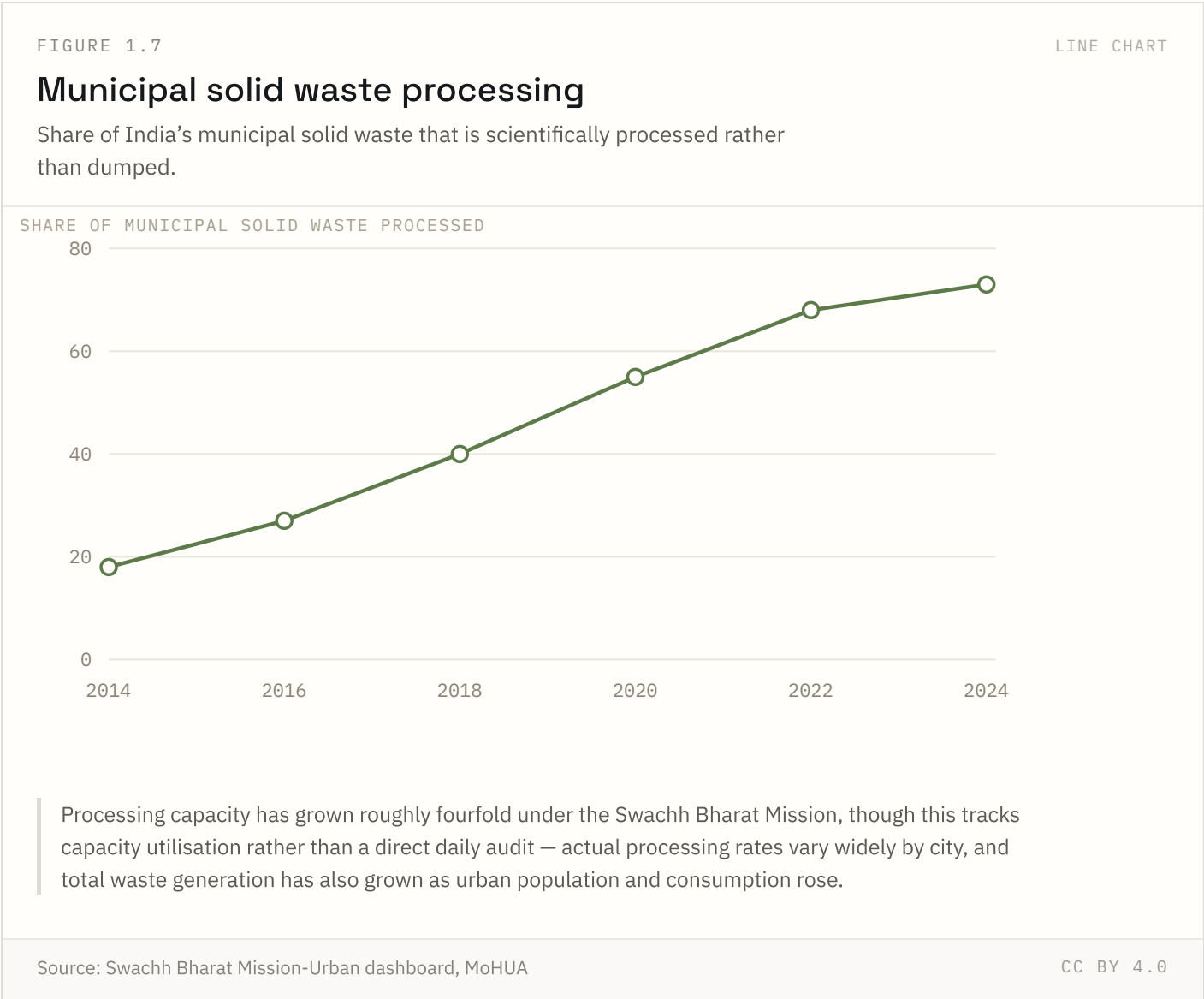


AMRUT 1.0 fell short of its full connection targets by its 2021 close; AMRUT 2.0 (launched 2021) aims at universal coverage. Sewerage lagged water supply throughout the scheme, reflecting the higher cost and longer build time of underground network projects.

Source: Atal Mission for Rejuvenation and Urban Transformation, MoHUA Coverage: 2015–2021 (AMRUT 1.0)

Updated: 18 Aug 2026

FIGURE 1.7 NATIONAL ONLY

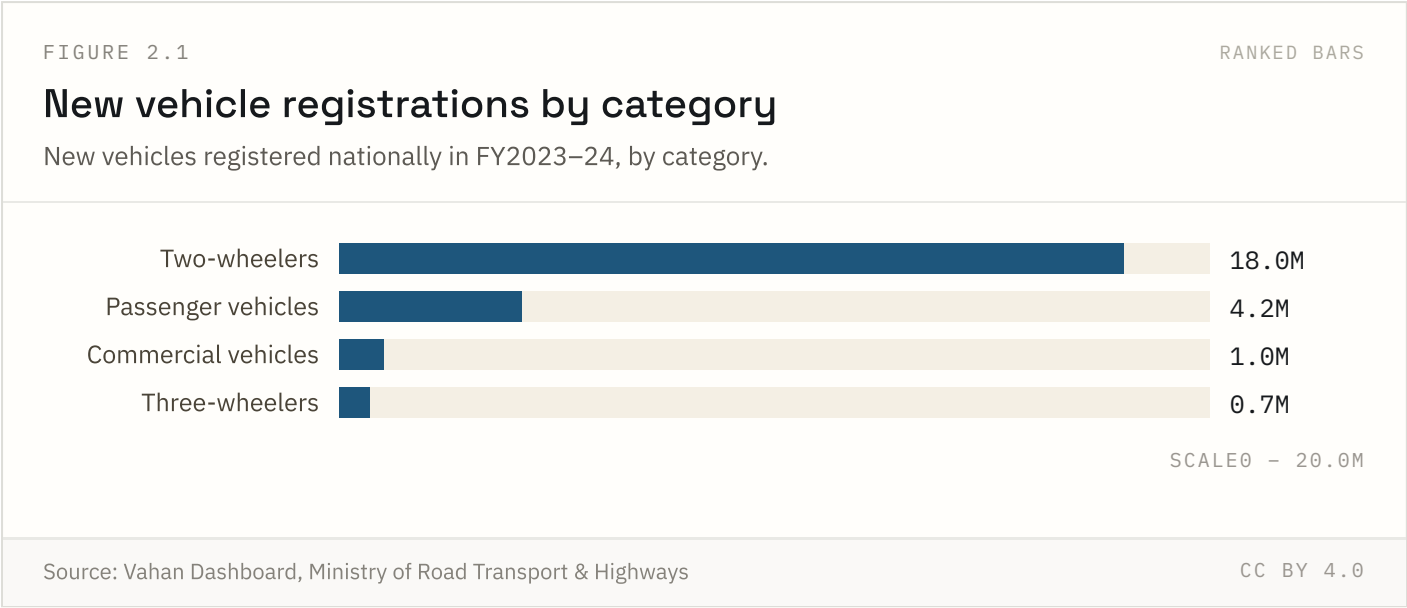


Processing capacity has grown roughly fourfold since 2014, but total waste generation has also risen with urban population and consumption growth — a rising processed share does not necessarily mean a falling absolute quantity dumped in landfills.

Transport & mobility

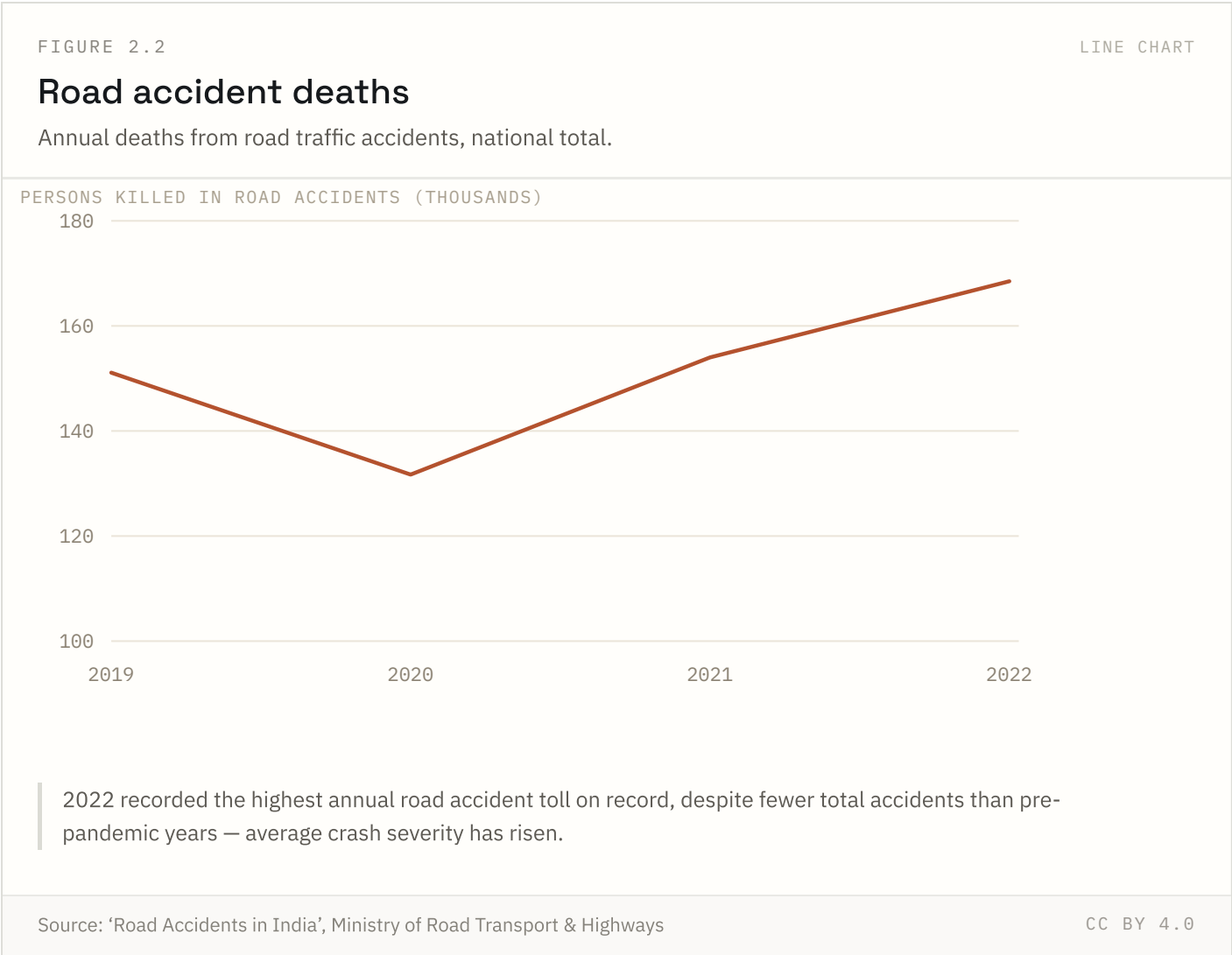
All 4 indicators in this chapter are published at national level only. No state breakdown is available from the sources used.

FIGURE 2.1 NATIONAL ONLY



Two-wheelers dominate India’s vehicle fleet by a wide margin — more than four times passenger vehicle registrations. Vahan records registrations, not sales, so figures can lag manufacturer-reported sales by weeks.

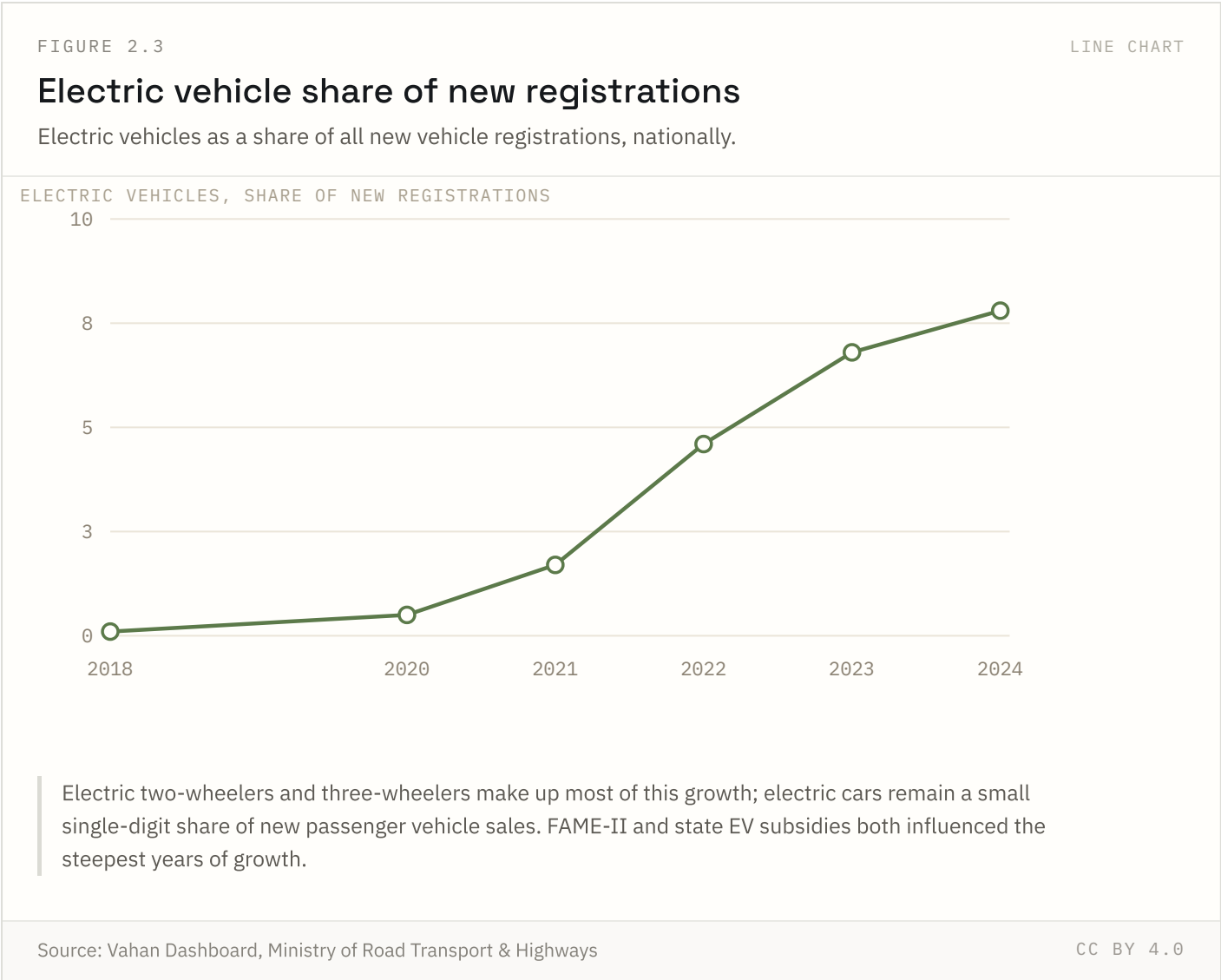
FIGURE 2.2 NATIONAL ONLY



2022 was the deadliest year on record even though reported accident counts were below 2019 levels — severity per crash has risen, partly reflecting more high-speed corridors. State reporting definitions vary, which limits comparability across states more than across years nationally.

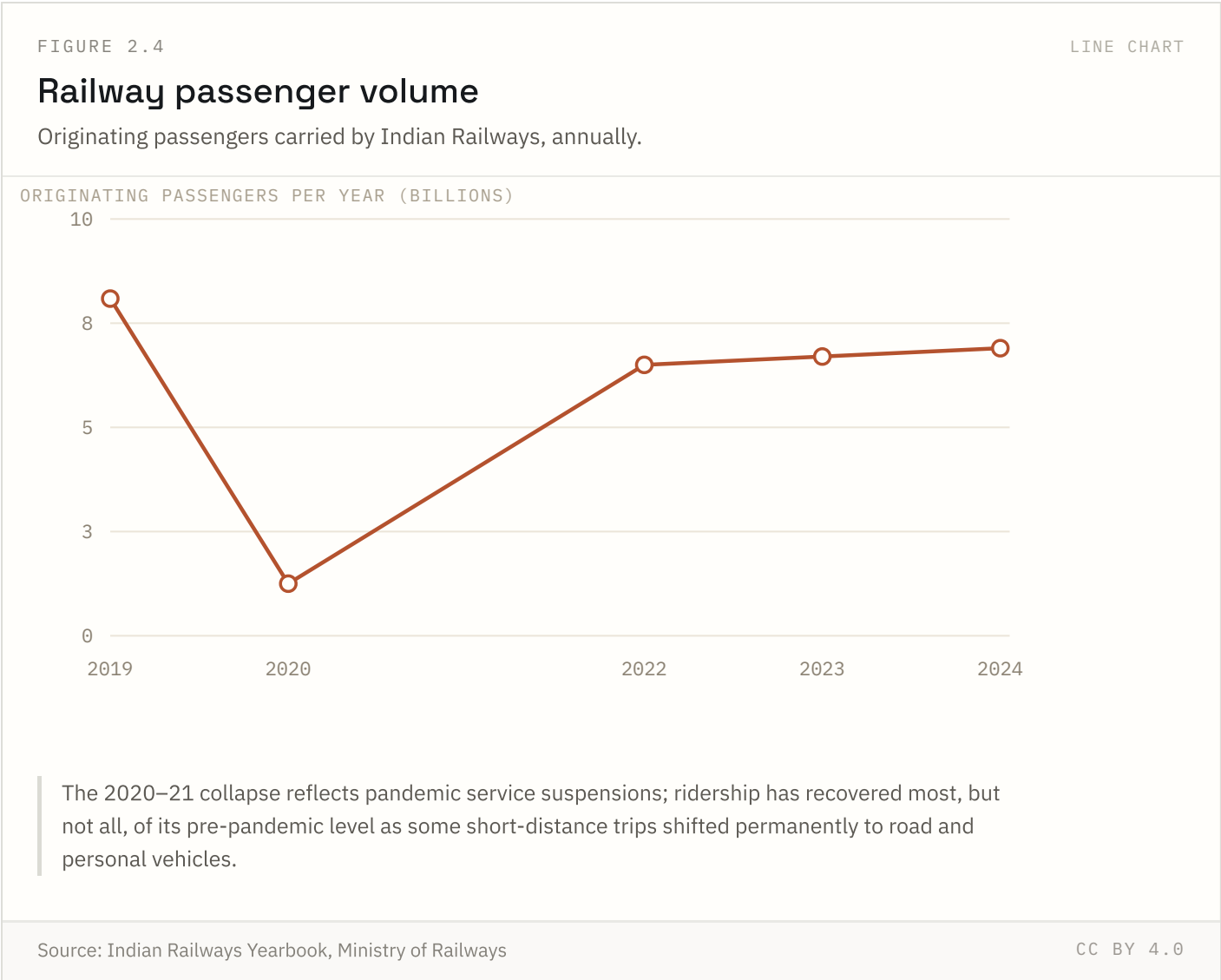
Source: 'Road Accidents in India', Ministry of Road Transport & Highways Coverage: 2019–2022

Updated: 18 Aug 2026



Electric two- and three-wheelers drive nearly all of this growth; electric passenger cars remain a small single-digit share of new car sales. Growth accelerated sharply after FAME-II and state-level EV subsidies took effect.

FIGURE 2.4 NATIONAL ONLY

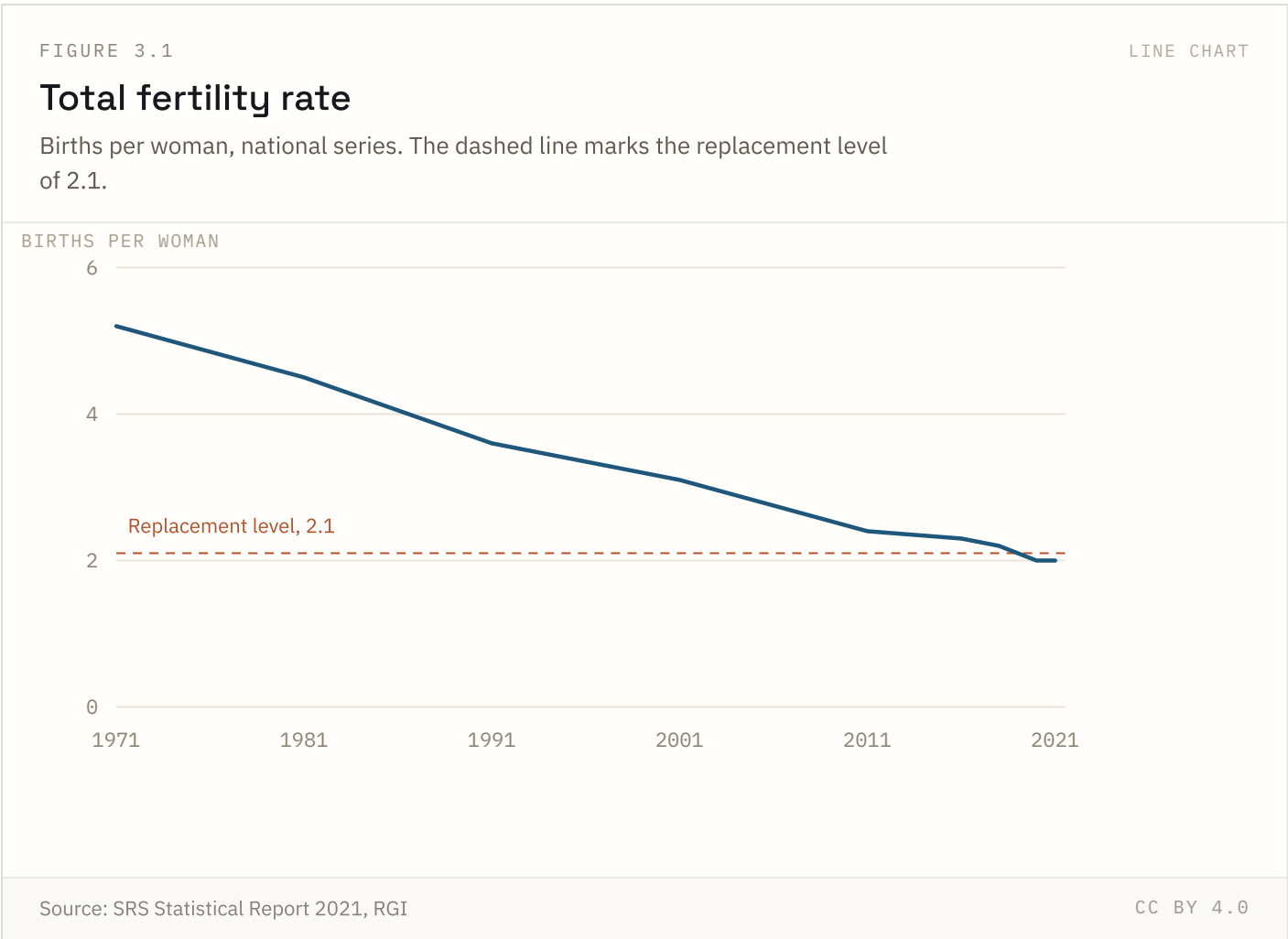


The 2020–21 collapse reflects pandemic service suspensions. Ridership has recovered most, though not all, of its pre-pandemic level — some short-distance trips appear to have shifted permanently to road and personal vehicles.

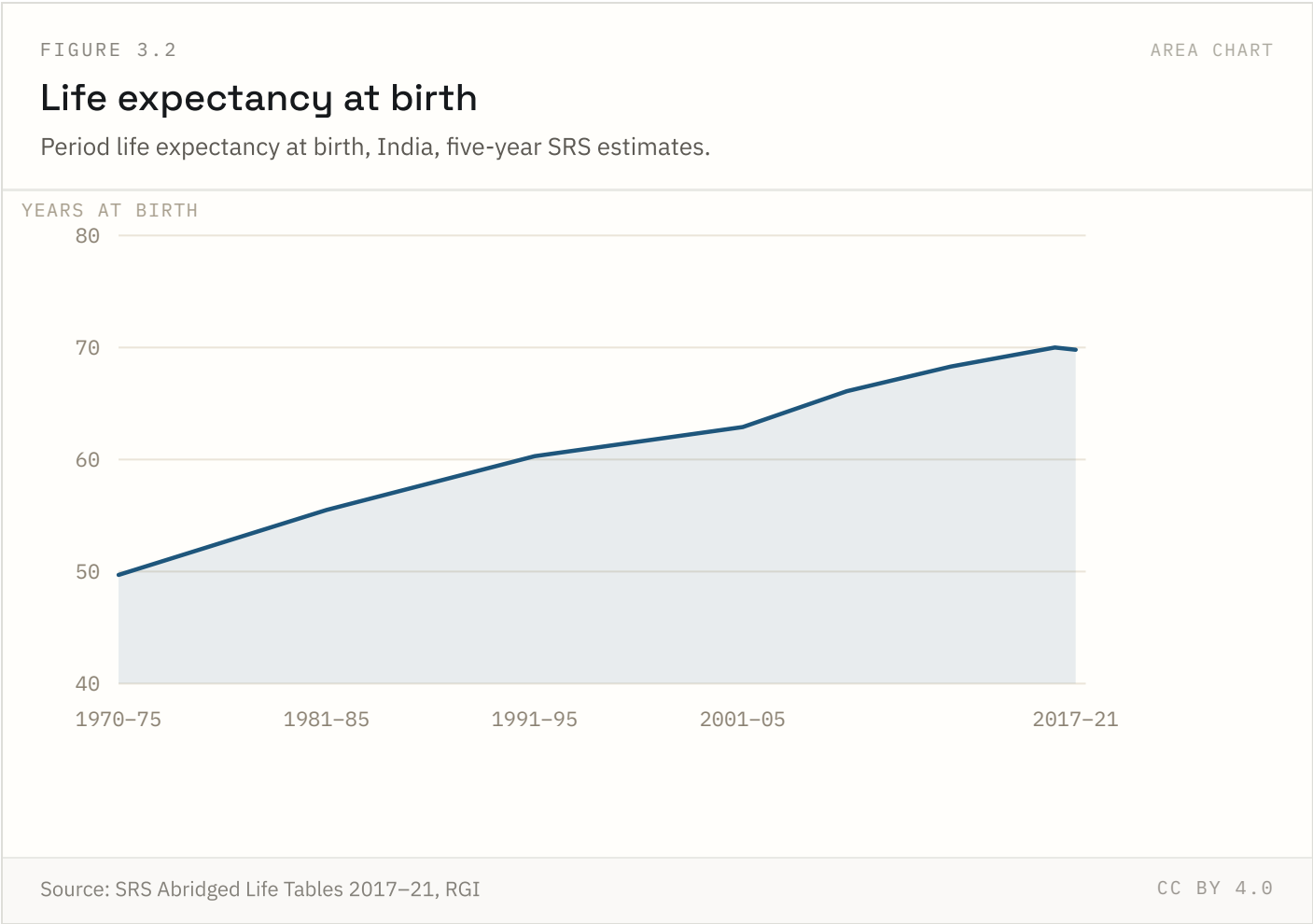
Population & demography

4 of 10 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 3.1 NATIONAL + STATE



The total fertility rate is the number of children a woman would bear if she experienced current age-specific fertility rates throughout her reproductive life. We splice decennial Census-based estimates to 1991 with the annual Sample Registration System thereafter; NFHS rounds are shown where they differ by more than 0.1.



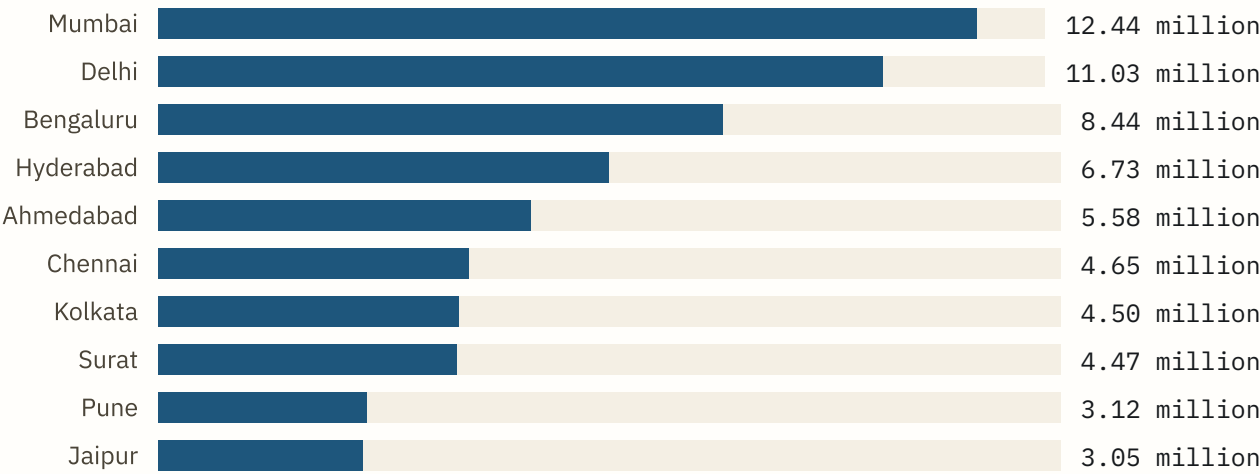
Period life expectancy at birth assumes a newborn experiences current mortality rates at every age for life. It is not a forecast of any individual cohort. The 2021 fall is real and reflects pandemic excess mortality; estimates of its size still vary across sources by several years.

FIGURE 3.3

RANKED BARS

Ten largest cities by population

Population within municipal limits (city proper), excluding the wider urban agglomeration.



SCALE 0 - 13.50 MILLION

City-proper population, within municipal limits — smaller than the wider urban agglomeration. Delhi's urban agglomeration (16.3 million) is roughly 48% larger than the city figure shown here.

Source: Census of India 2011

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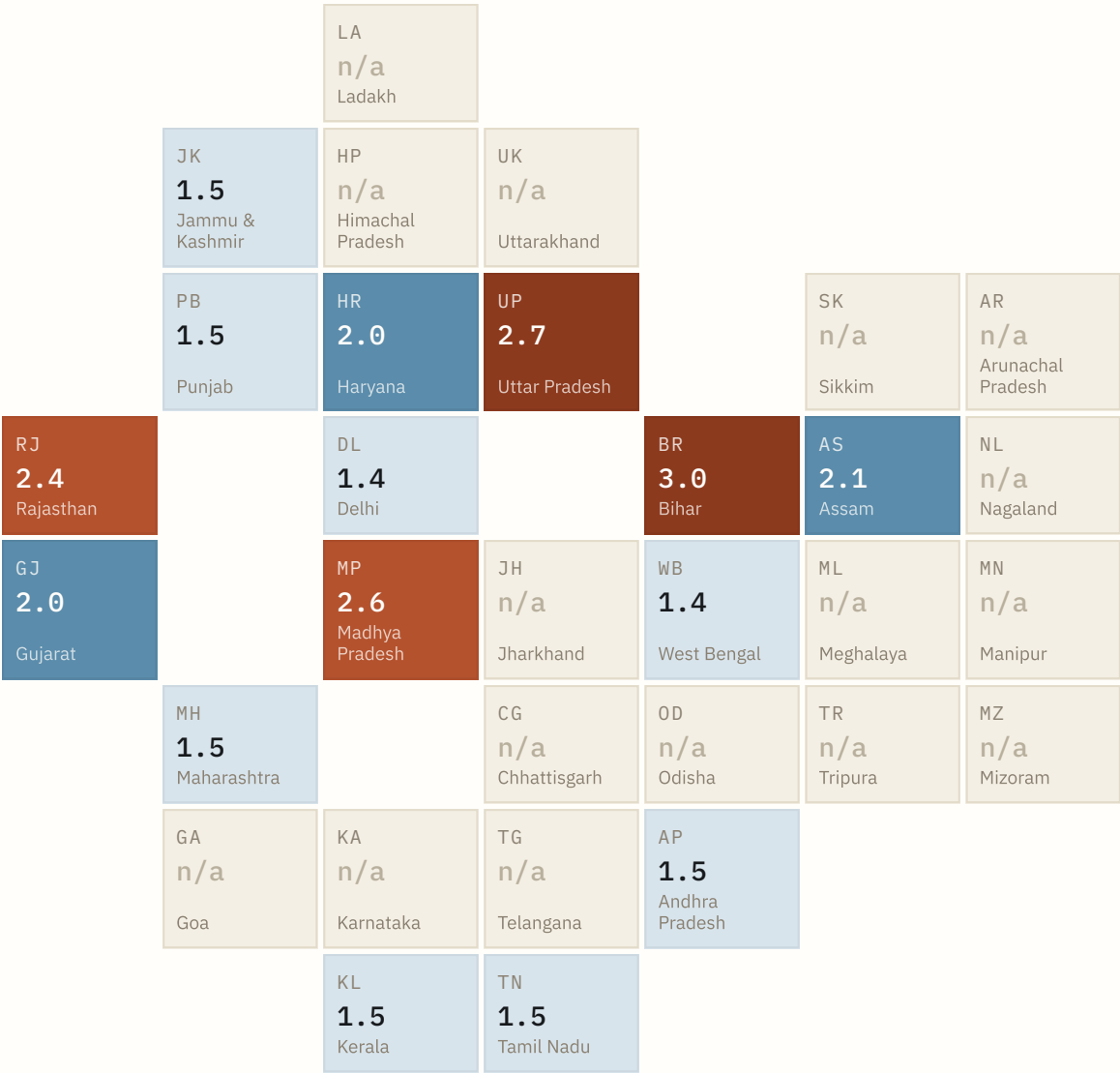
City-proper population excludes adjoining outgrowths and satellite towns, so it understates real metropolitan size — Delhi's urban agglomeration (16.3 million) is roughly 48% larger than its city-proper figure. We use city-proper because it is the more consistently defined boundary across cities.

FIGURE 3.4

TILE MAP

Fertility rate — tile map

One equal-area tile per State and UT, positioned to approximate relative geography.



≤ 1.5 1.6 – 1.9 2.0 – 2.2 2.3 – 2.6 ≥ 2.7

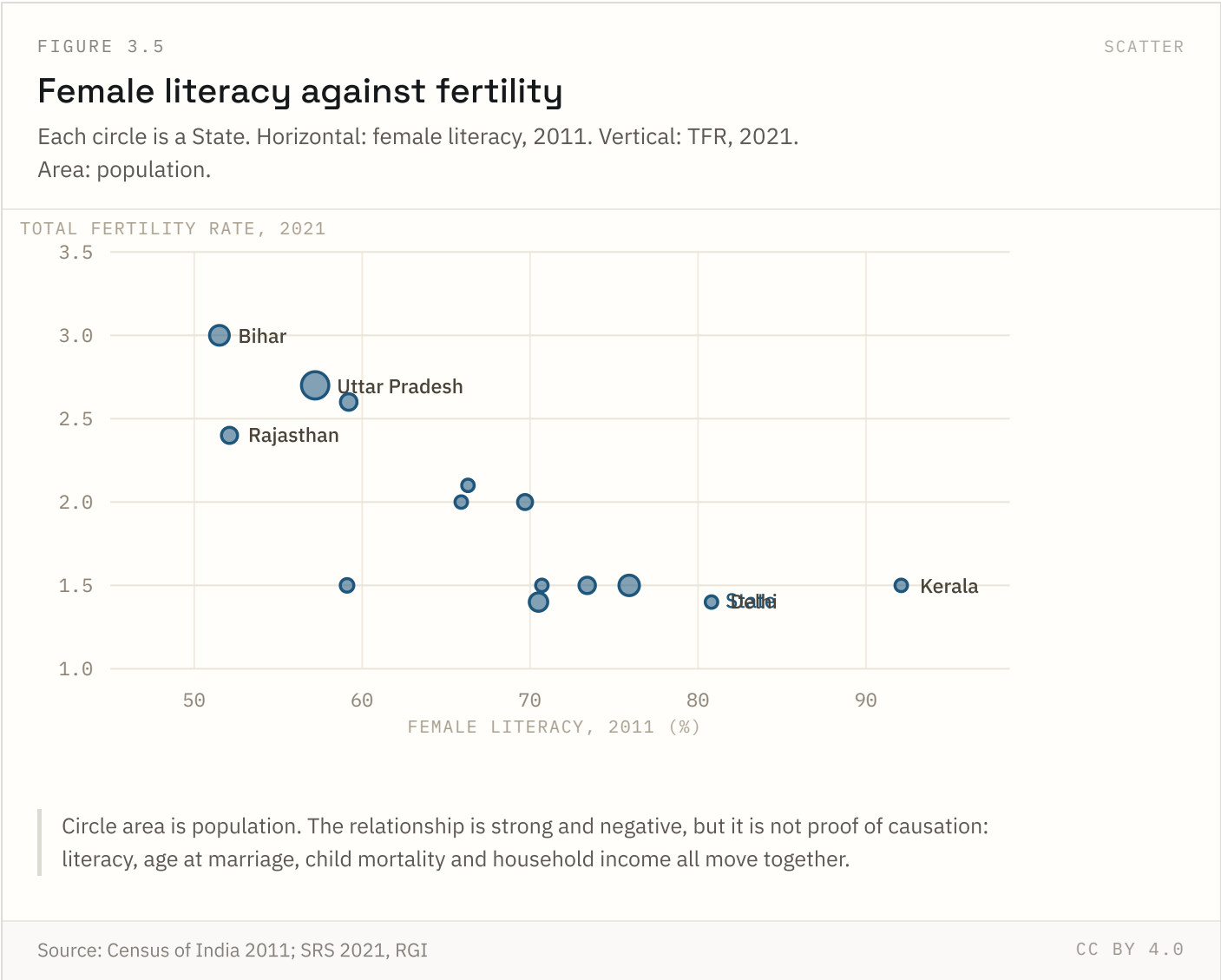
Schematic tile map: one tile per State / UT, positioned to approximate relative geography. Tiles are equal-area, so no state is visually over-weighted by land mass.

Source: SRS Statistical Report 2021, RGI

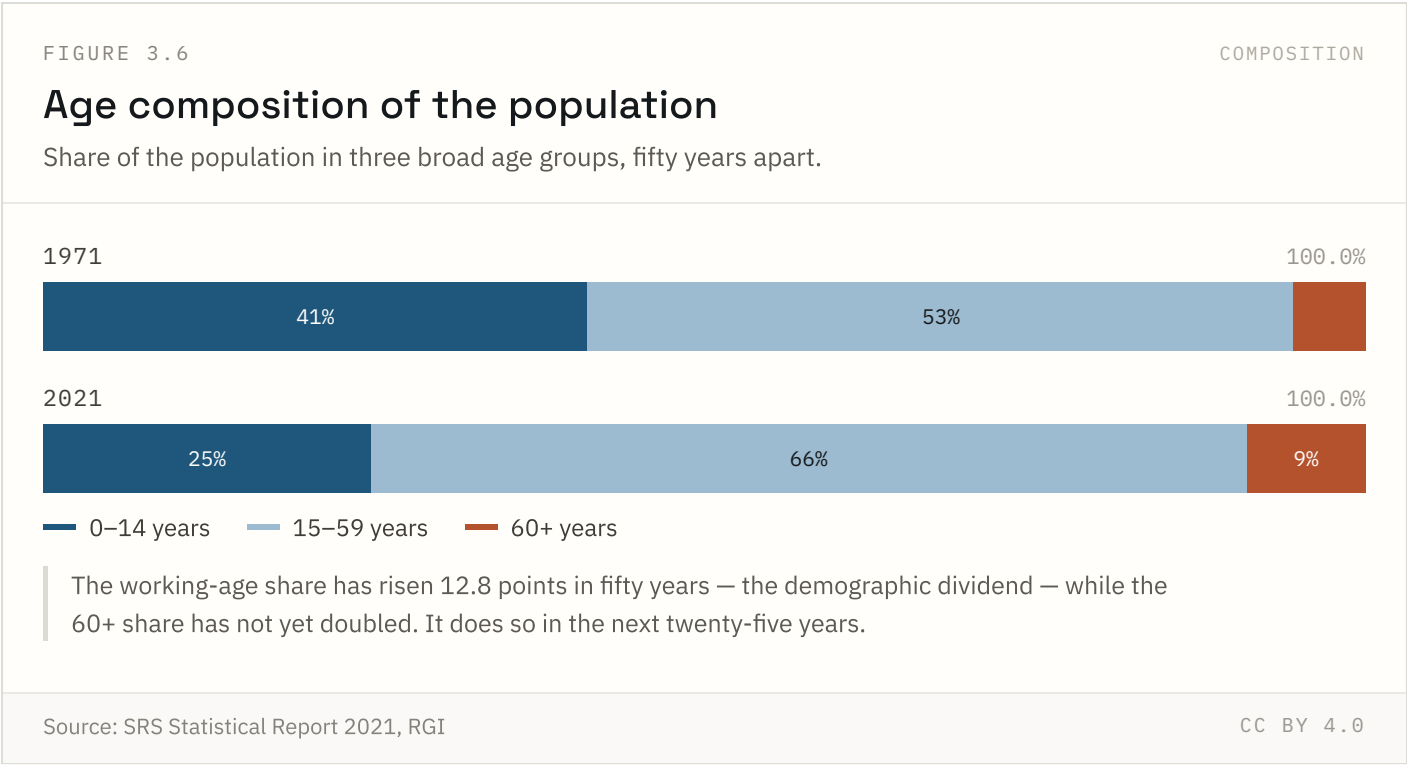
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A tile map removes the visual bias of land area: Rajasthan does not shout louder than Kerala. Tiles marked n/a are States and UTs for which the SRS 2021 report does not publish a separate TFR estimate.

FIGURE 3.5 NATIONAL + STATE

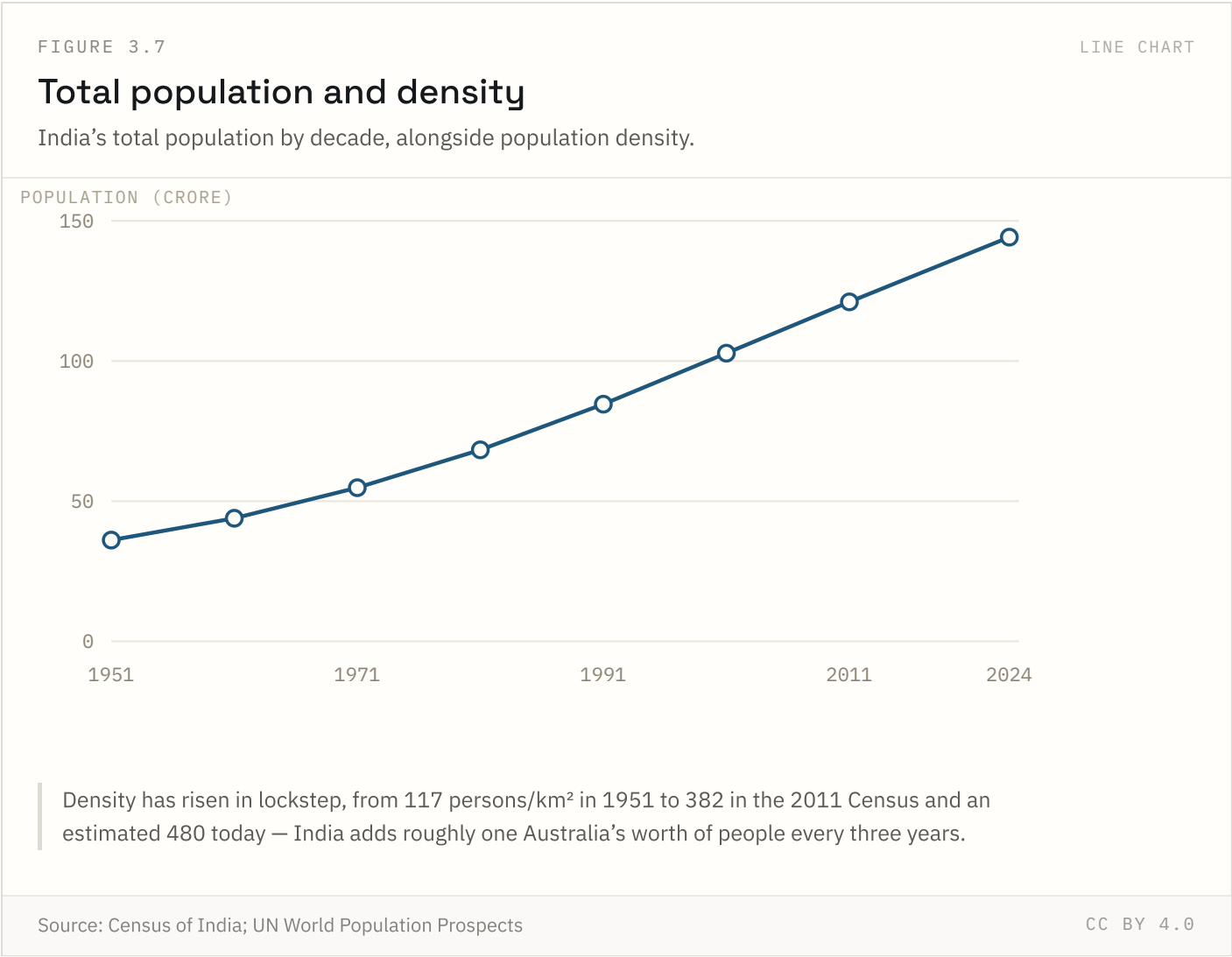


The negative relationship is strong across States, but it is not evidence of causation on its own — literacy, age at marriage, child mortality and household income all move together. Read it as a description of the fertility map, not a mechanism.



The 0–14 share fell from 41.2% to 24.8% between 1971 and 2021 while the working-age share rose to 66.2%. That gap is the demographic dividend; it closes as the 60+ share, now 9.0%, roughly doubles by 2050.

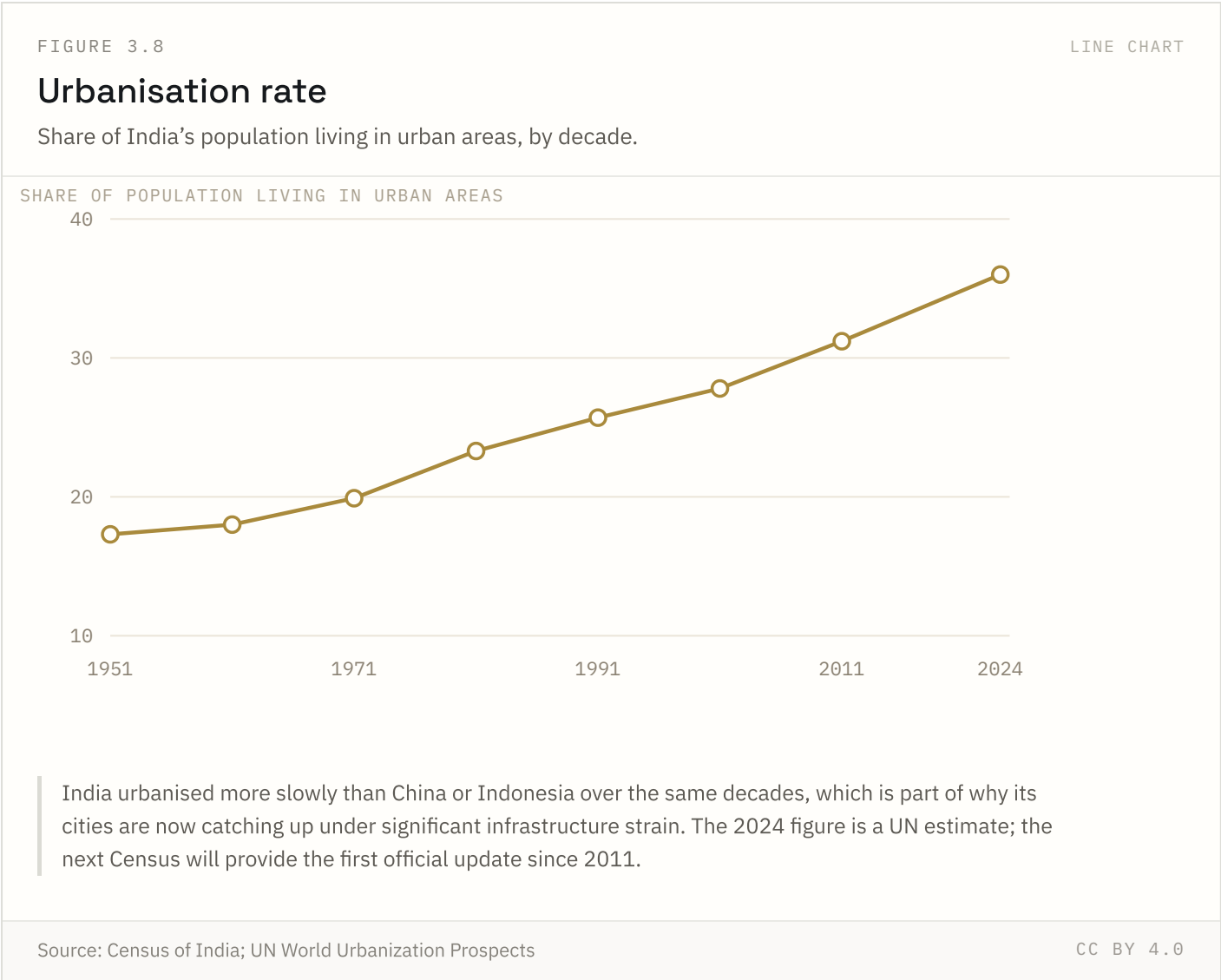
FIGURE 3.7 NATIONAL ONLY



India crossed 1 billion people around 2000 and an estimated 1.44 billion by 2024, having overtaken China as the world's most populous country in 2023. Density figures assume a fixed land area, so they track population growth almost exactly.

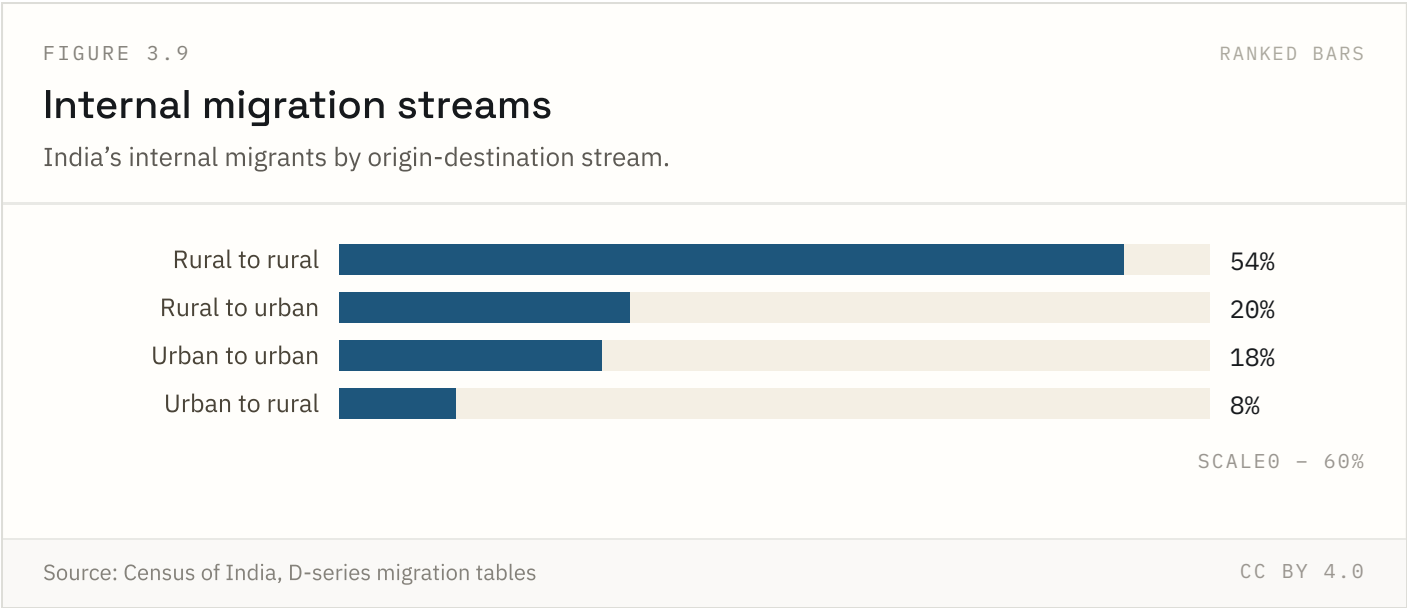
Source: Census of India; UN World Population Prospects Coverage: 1951–2024 Updated: 18 Aug 2026

FIGURE 3.8 NATIONAL ONLY



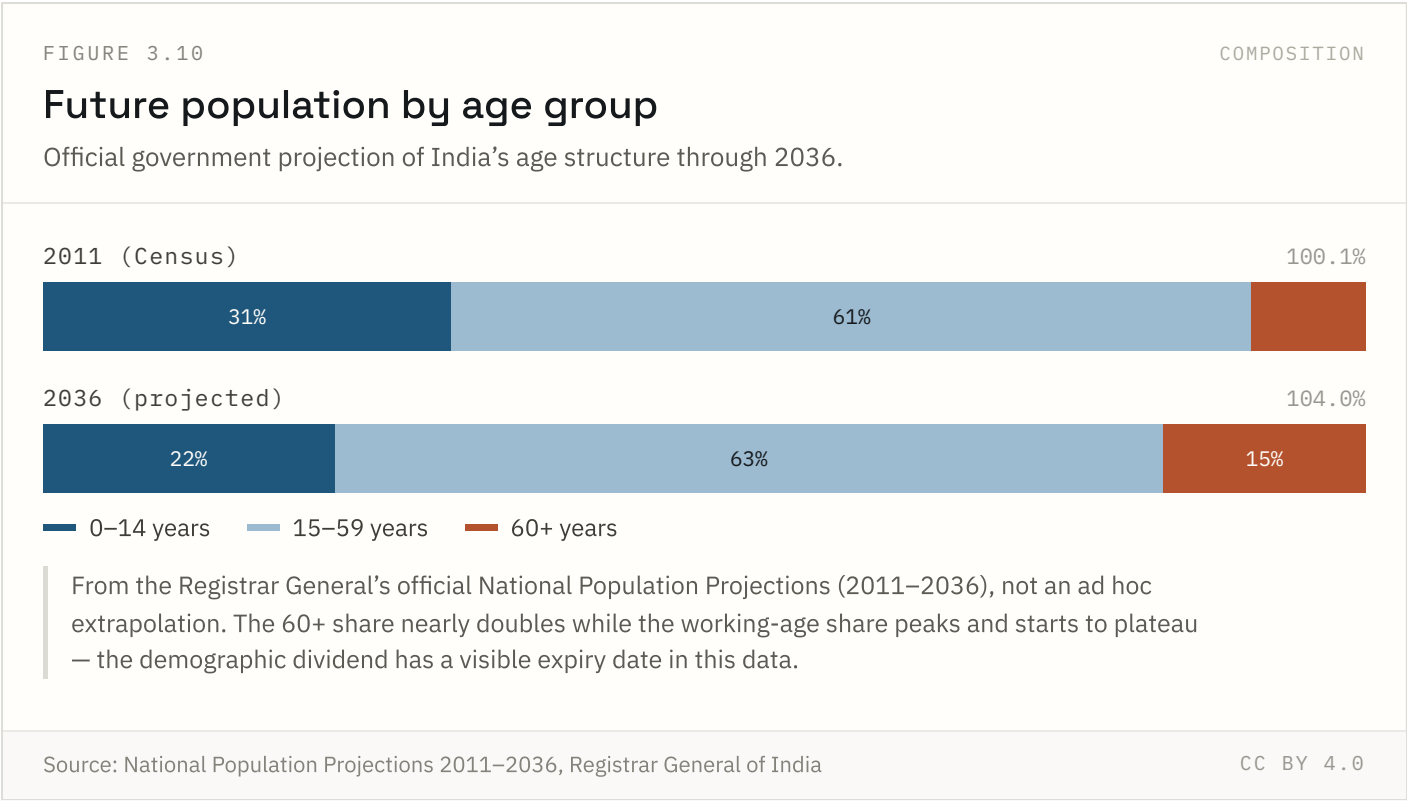
India’s urbanisation has been notably gradual compared to East Asian peers — a mixed blessing that avoided some of the informal-settlement crises seen elsewhere, but also means the next wave of urban growth, now underway, is arriving on infrastructure built for a smaller urban population.

FIGURE 3.9 NATIONAL ONLY



Rural-to-rural migration, mostly marriage-related moves by women, is the largest stream by volume — rural-to-urban labour migration, the stream most associated with India’s growth story, is actually the second largest, not the biggest.

Source: Census of India, D-series migration tables Coverage: 2011 Updated: 18 Aug 2026



This is RGI's own published projection, not an extrapolation built for this site. It shows the demographic dividend has a visible expiry date: the working-age share is projected to plateau around the mid-2030s as the 60+ share nearly doubles from its 2011 level.

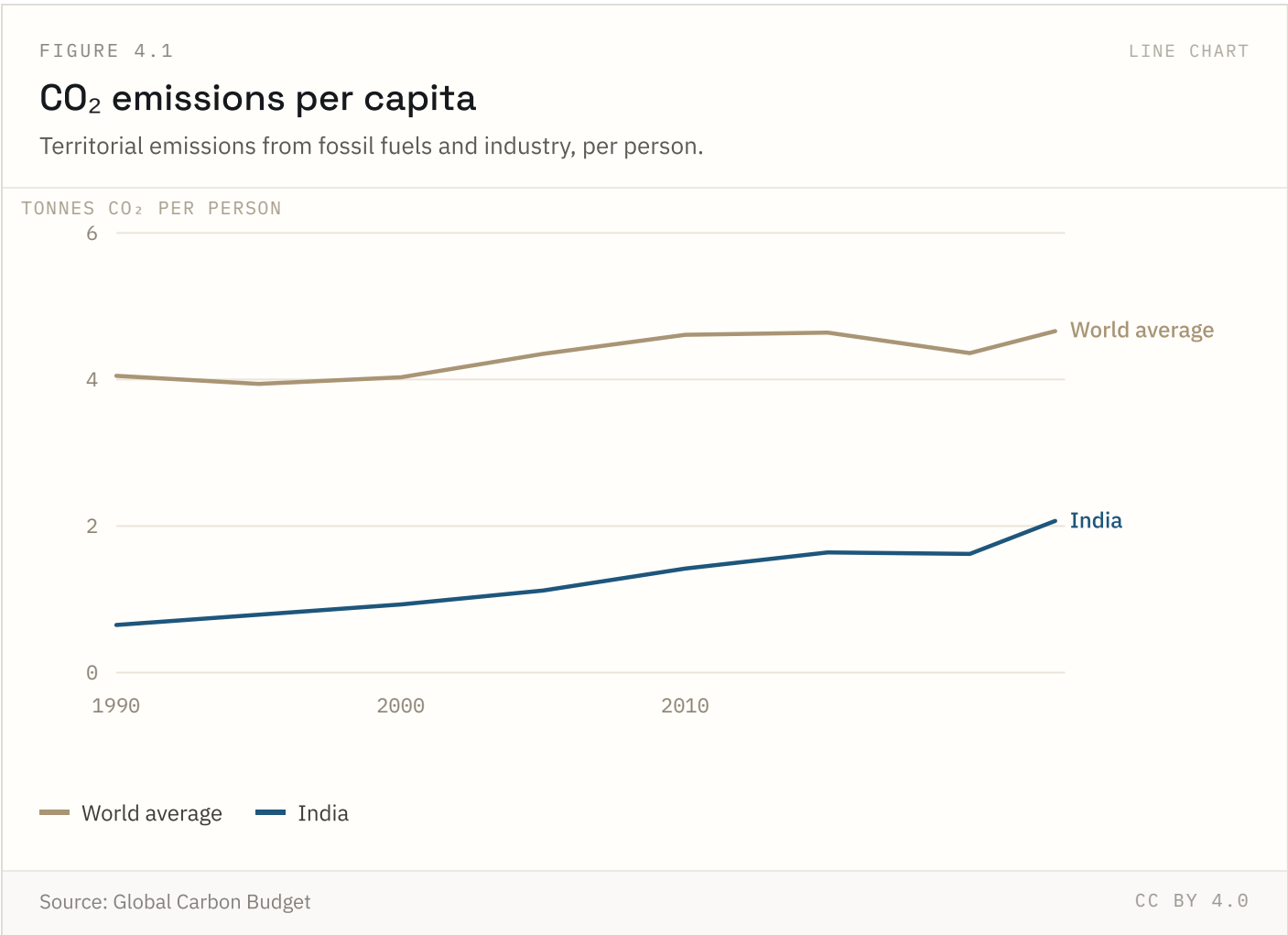
Source: National Population Projections 2011–2036, Registrar General of India

Coverage: 2011 to 2036 (projected) Updated: 18 Aug 2026

Climate & air quality

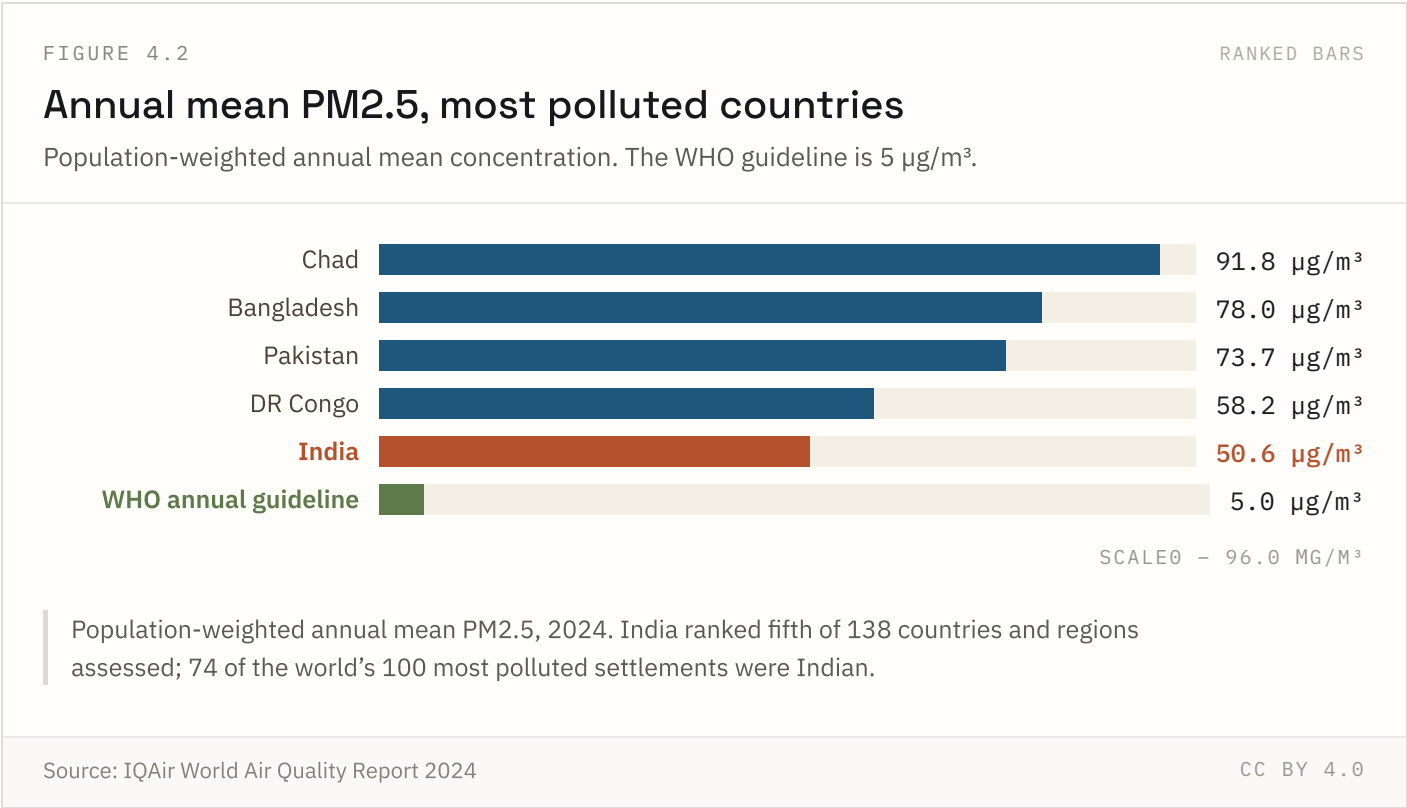
1 of 10 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 4.1 NATIONAL ONLY



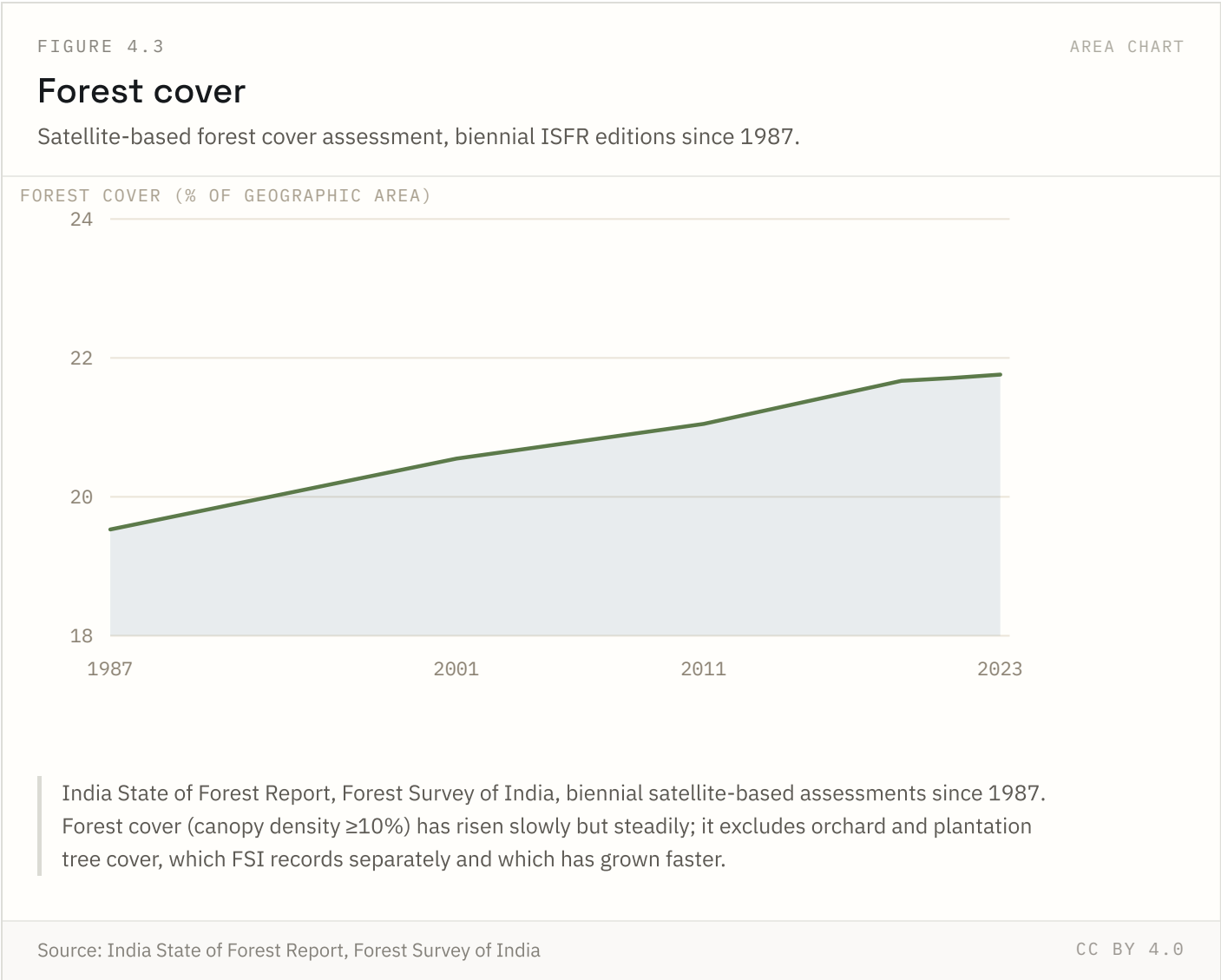
India is the third-largest emitter in absolute terms and among the lowest per person of any major economy — roughly 45 per cent of the world average. Cumulative emissions since 1850 are about 3 per cent of the global total.

FIGURE 4.2 NATIONAL ONLY



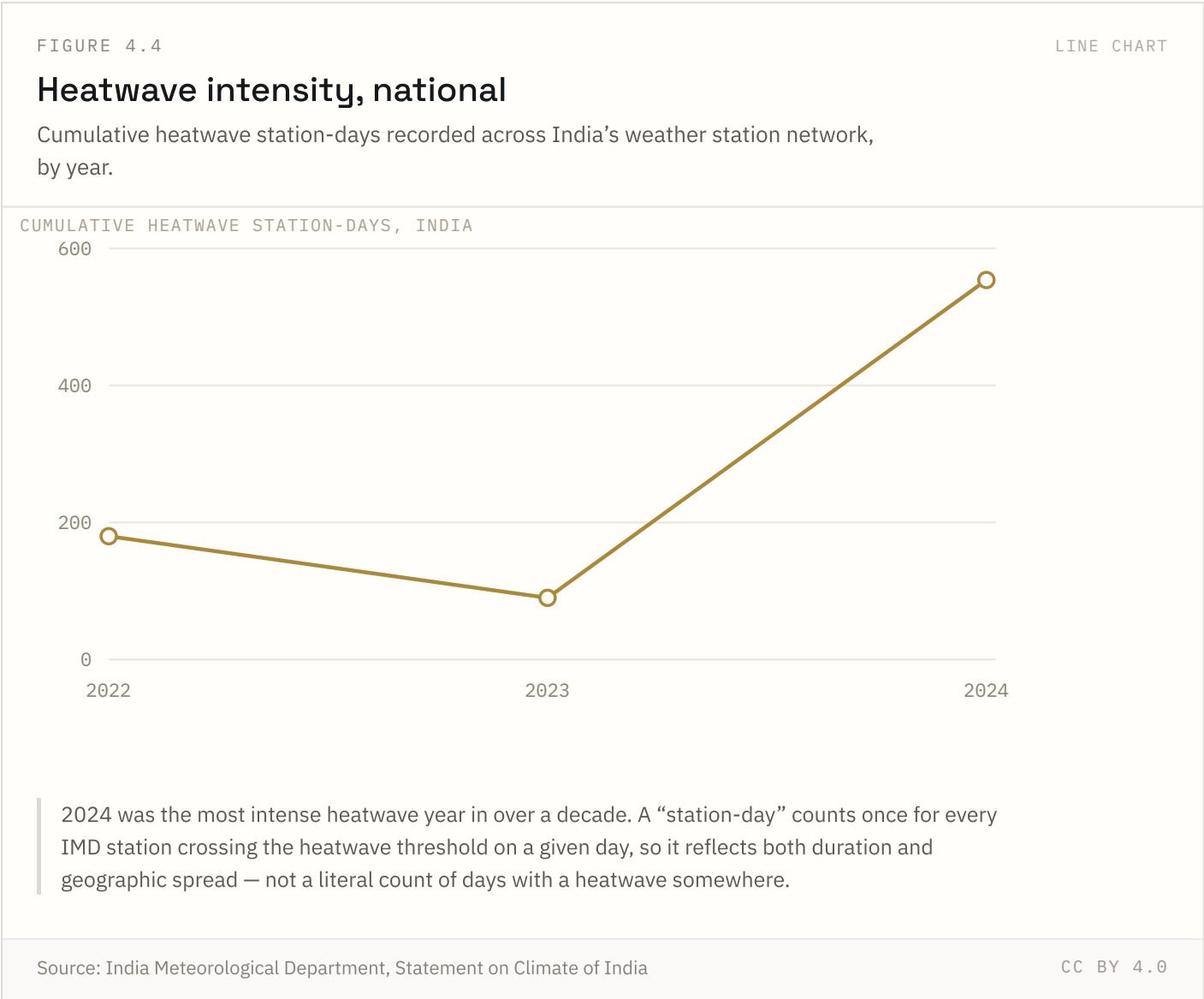
Every city shown exceeds the WHO annual guideline of 5 µg/m³, and all but one exceed India’s own standard of 40. Monitor coverage is uneven: much of central and eastern India has fewer than one continuous monitor per million people.

FIGURE 4.3 NATIONAL ONLY



Forest cover has risen slowly but steadily, from 19.53% of geographic area in 1987 to 21.76% in 2023 — still short of the National Forest Policy target of 33%. The figure excludes orchard and plantation tree cover, tracked separately, which has grown faster than natural forest.

FIGURE 4.4 NATIONAL ONLY

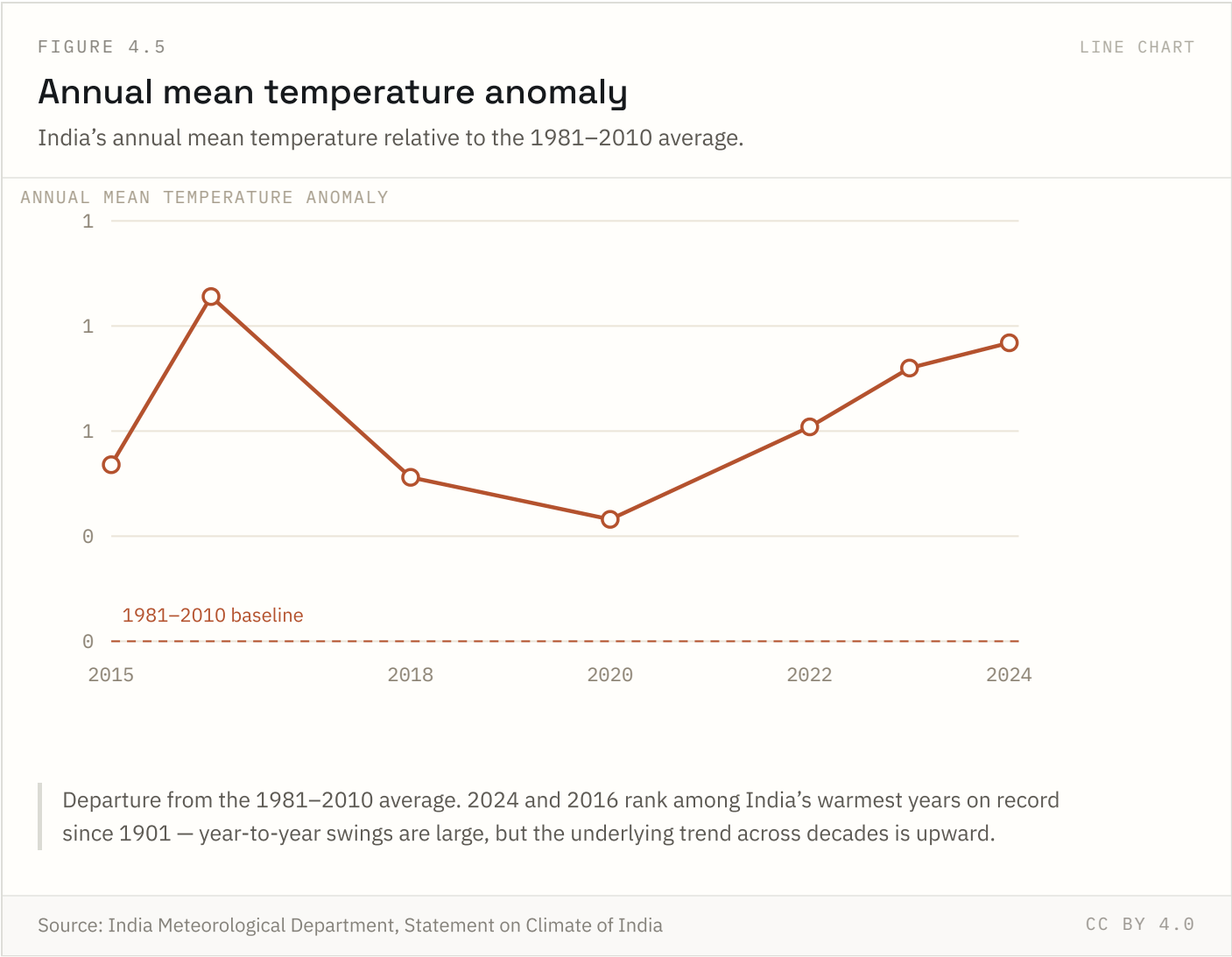


2024 was widely reported as India’s most intense heatwave year in over a decade, with several cities recording temperatures above 45°C. The station-day metric rewards both longer heatwaves and wider geographic spread, so a single very hot, very long event in a densely monitored region can dominate the national total.

Source: India Meteorological Department, Statement on Climate of India Coverage: 2022–2024

Updated: 18 Aug 2026

FIGURE 4.5 NATIONAL ONLY

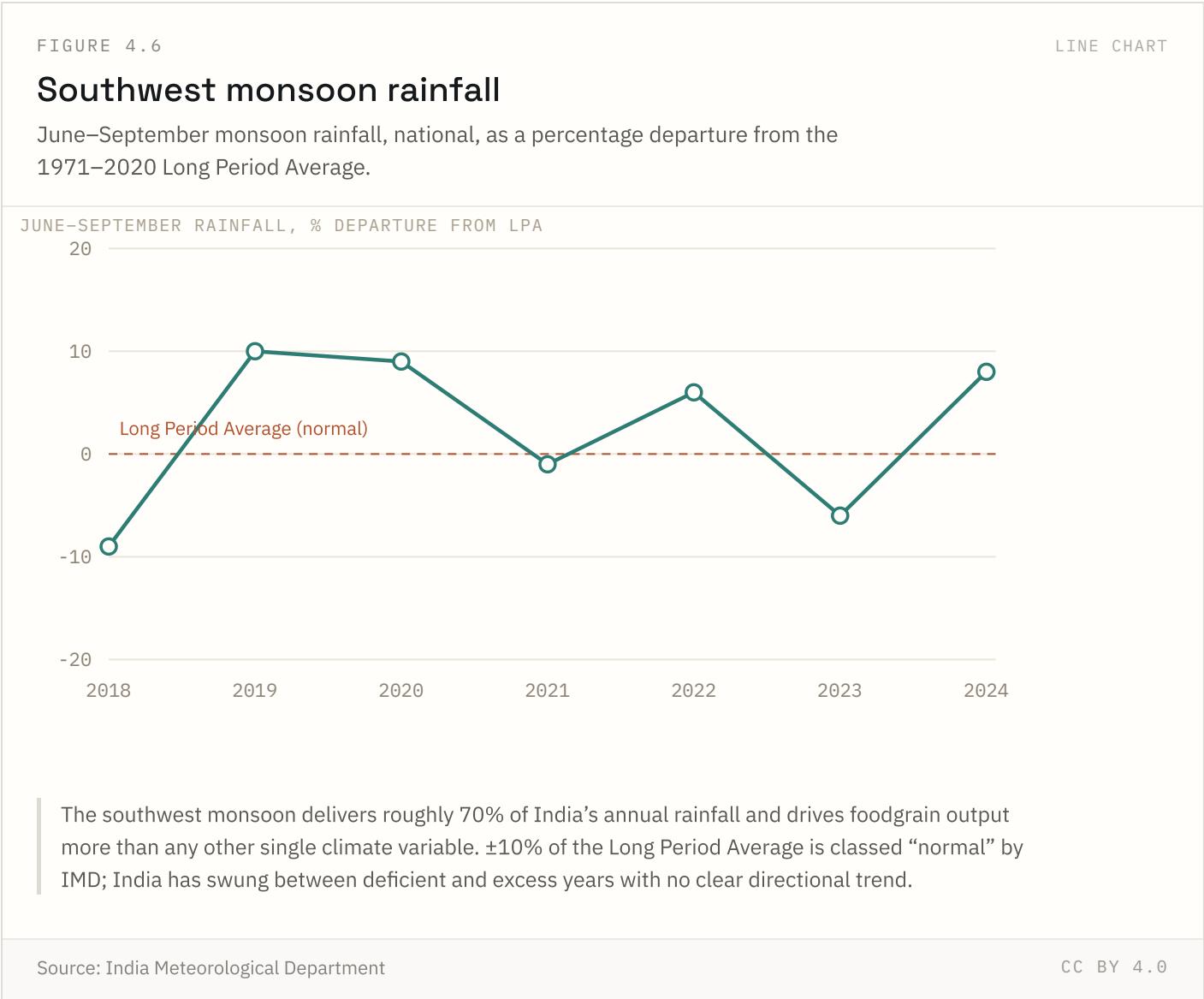


IMD has ranked several recent years among the warmest since national records began in 1901. Annual figures are noisy year to year — a single cool year does not reverse the multi-decade warming trend visible in the underlying data.

Source: India Meteorological Department, Statement on Climate of India Coverage: 2015–2024

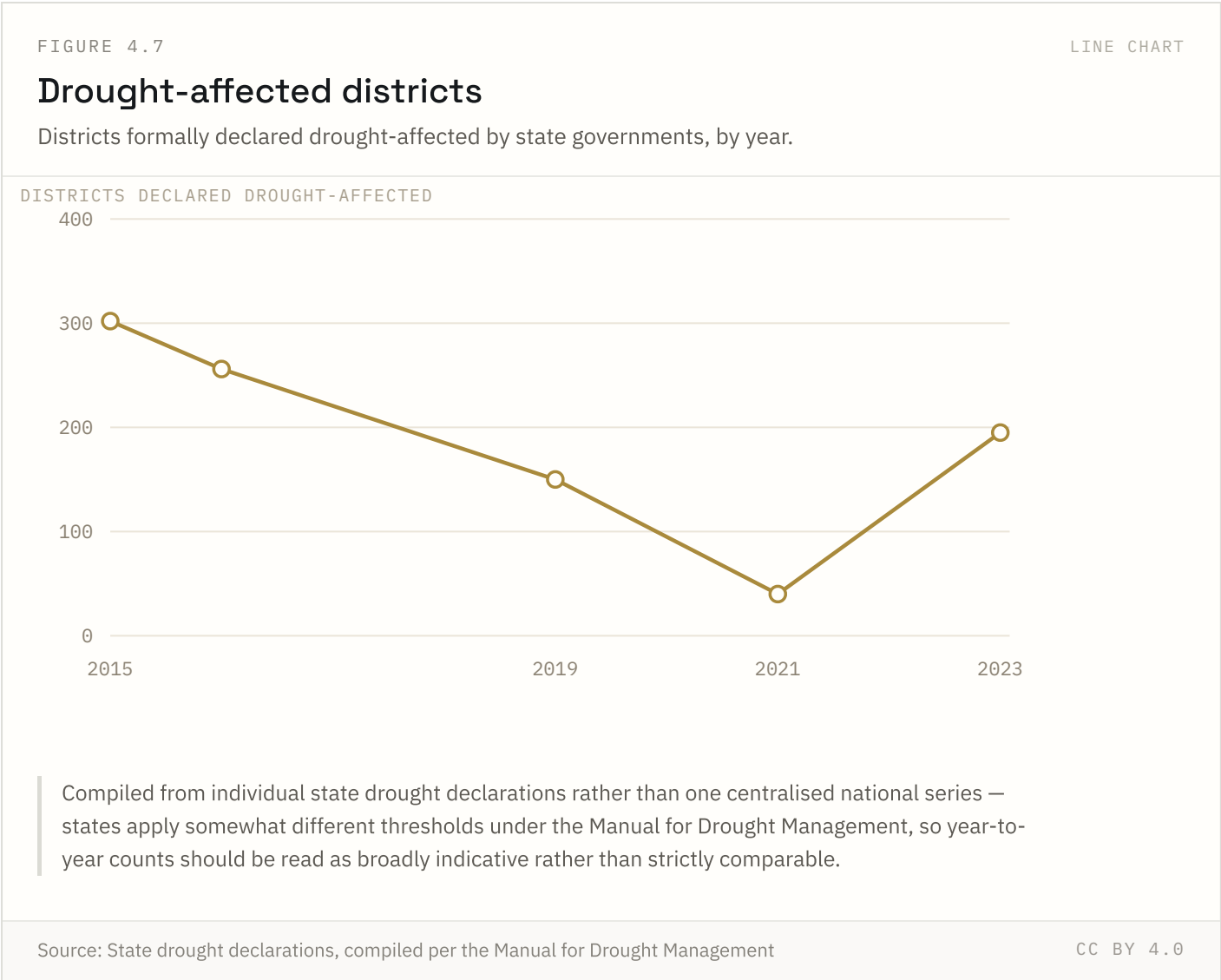
Updated: 18 Aug 2026

FIGURE 4.6 NATIONAL ONLY



IMD classes $\pm 10\%$ of the LPA as “normal.” The monsoon’s national total can mask sharply uneven regional distribution — a “normal” year on paper can still combine drought in one region with flooding in another.

FIGURE 4.7 NATIONAL ONLY

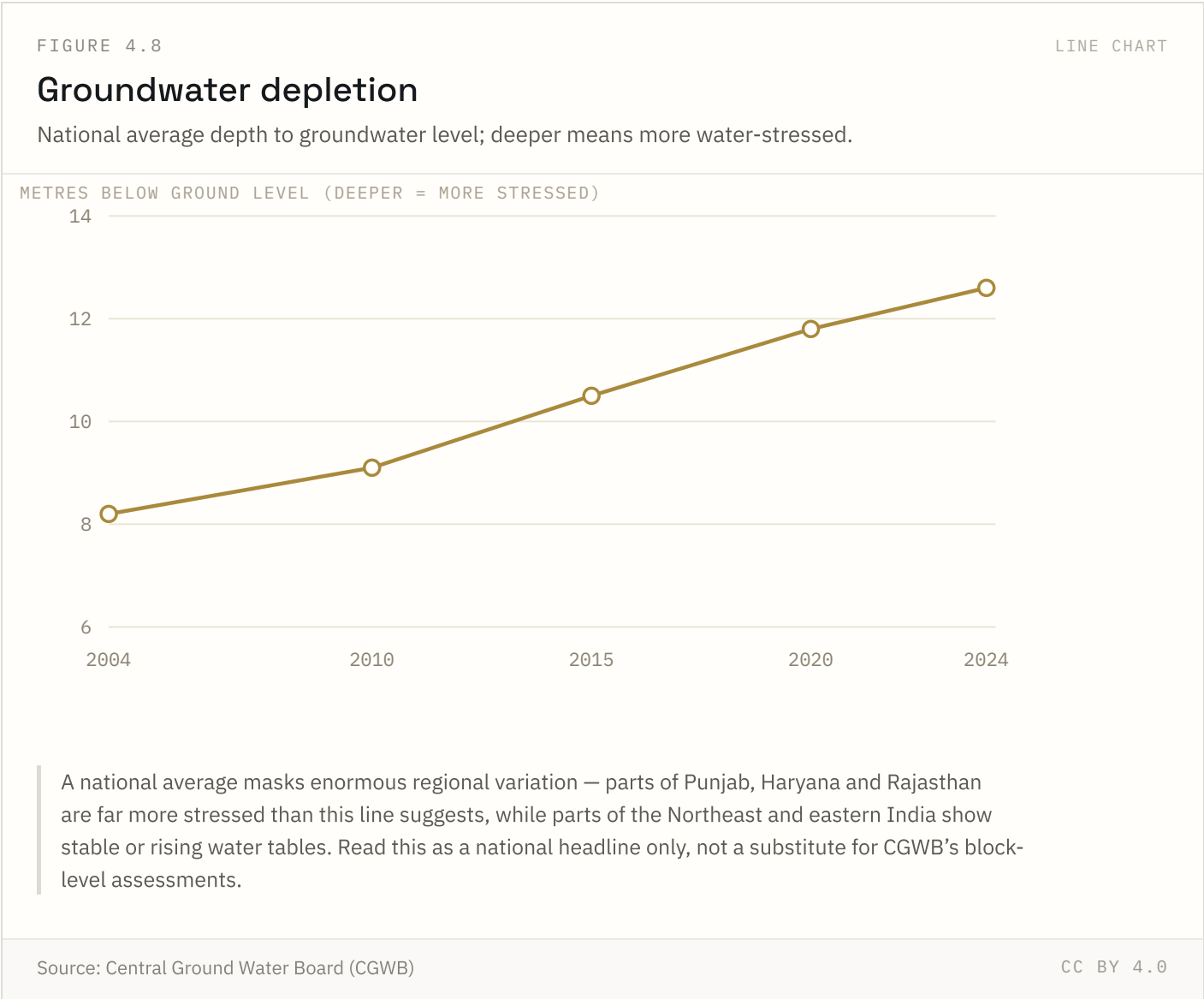


There is no single centralised national drought register — this compiles individual state declarations, which apply the Manual for Drought Management’s criteria with some local discretion. Treat trends as broadly indicative; exact counts vary by source and reporting date.

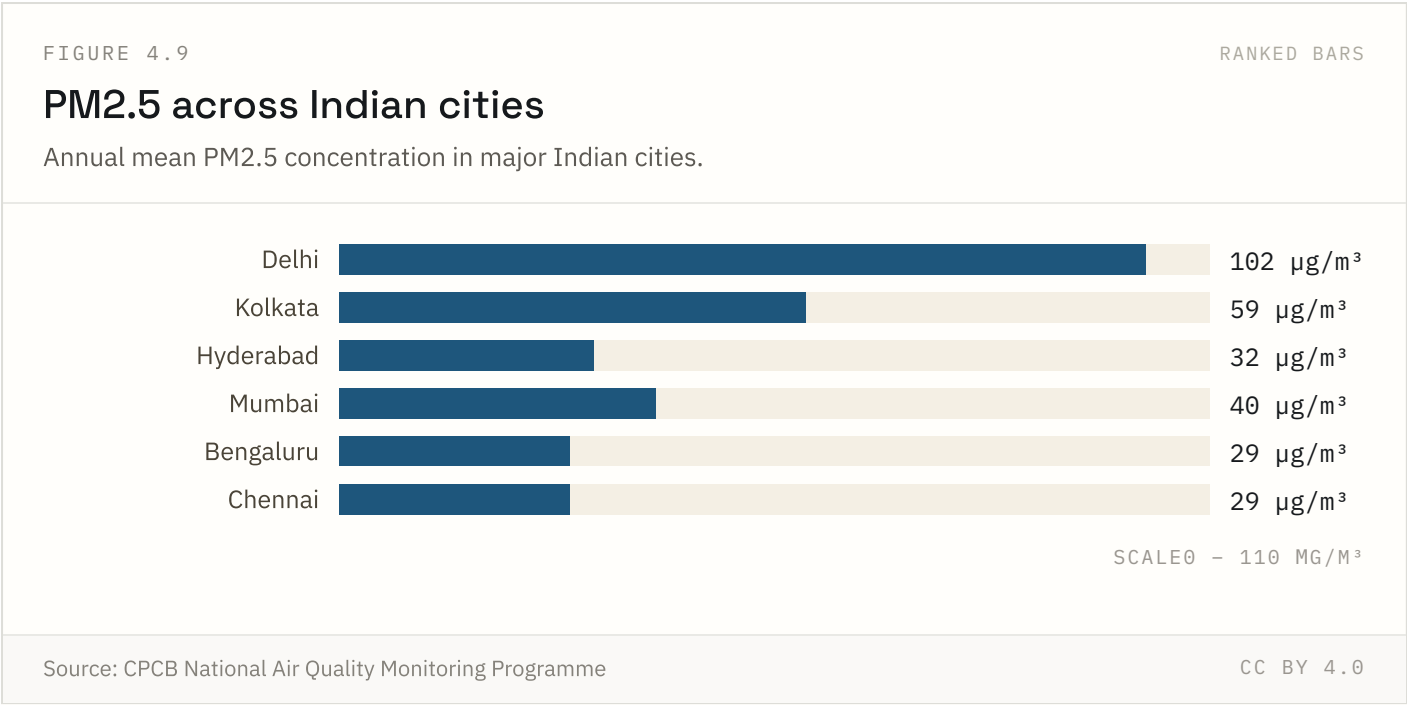
Source: State drought declarations, compiled per the Manual for Drought Management Coverage: 2015–2023

Updated: 18 Aug 2026

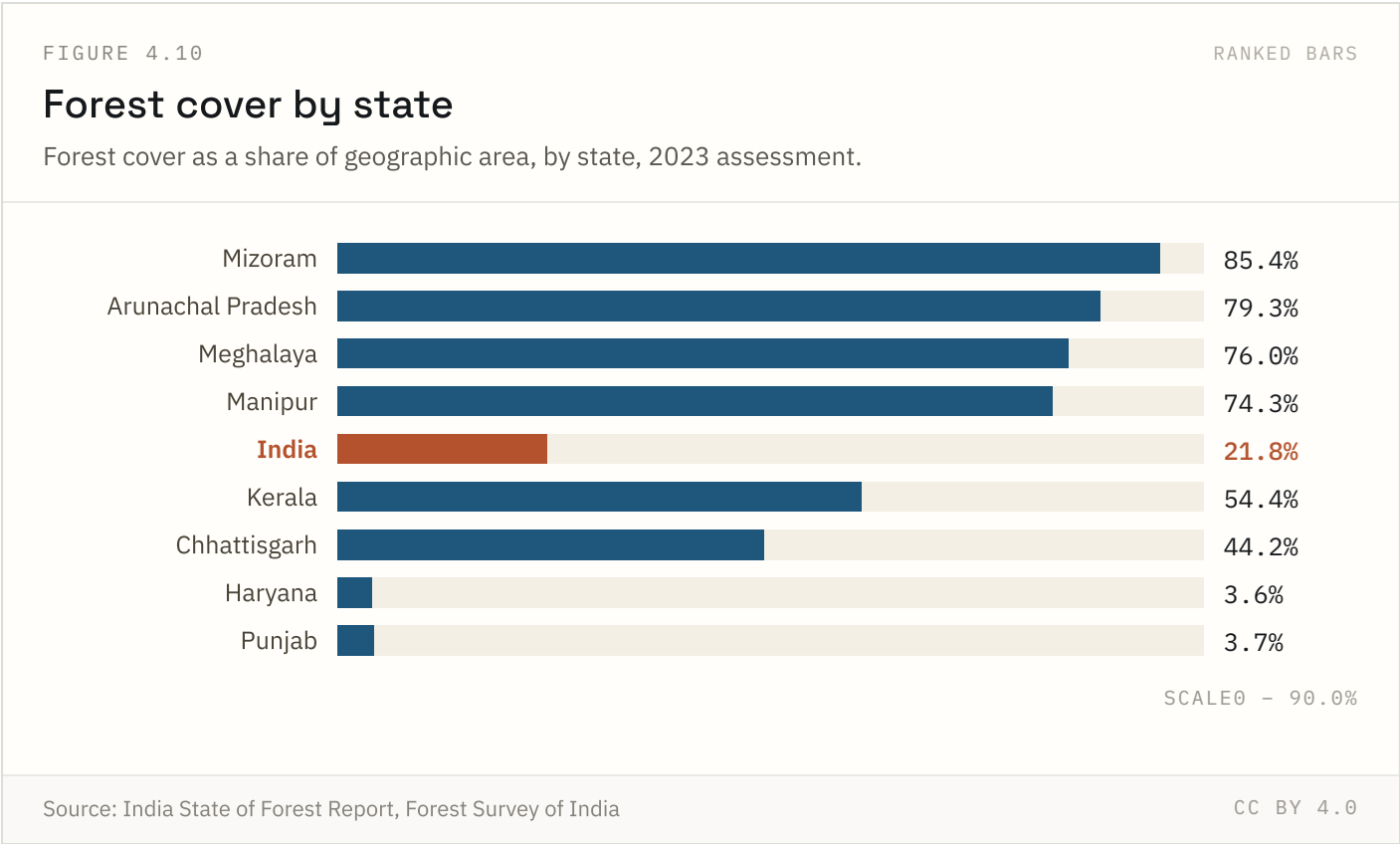
FIGURE 4.8 NATIONAL ONLY



India is the world’s largest extractor of groundwater, used mainly for irrigation. A national average conceals sharp regional divergence — Punjab, Haryana and Rajasthan are critically over-extracted, while parts of eastern India show stable or even rising water tables.



Every city shown exceeds both the WHO guideline (5 $\mu\text{g}/\text{m}^3$) and India’s own national standard (40 $\mu\text{g}/\text{m}^3$). Delhi’s winter inversion and regional stubble-burning make its annual average an outlier even among India’s other large, heavily polluted cities.



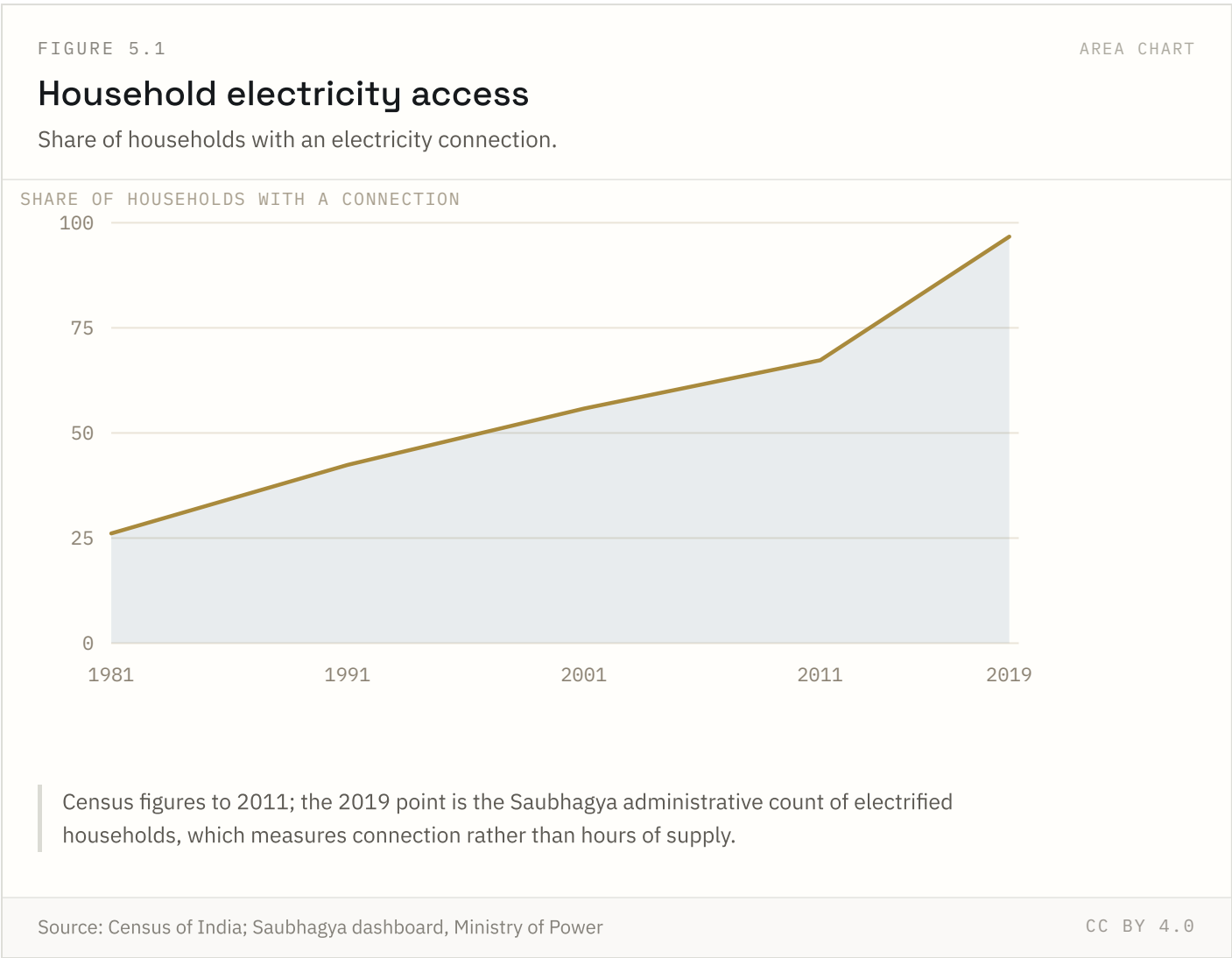
The gap between India’s most and least forested major states spans more than 80 percentage points. Northeastern states carry a hugely disproportionate share of India’s forest cover relative to their land area’s share of the country.

Highest: **Mizoram** at 85.4%. Lowest: **Haryana** at 3.6%.

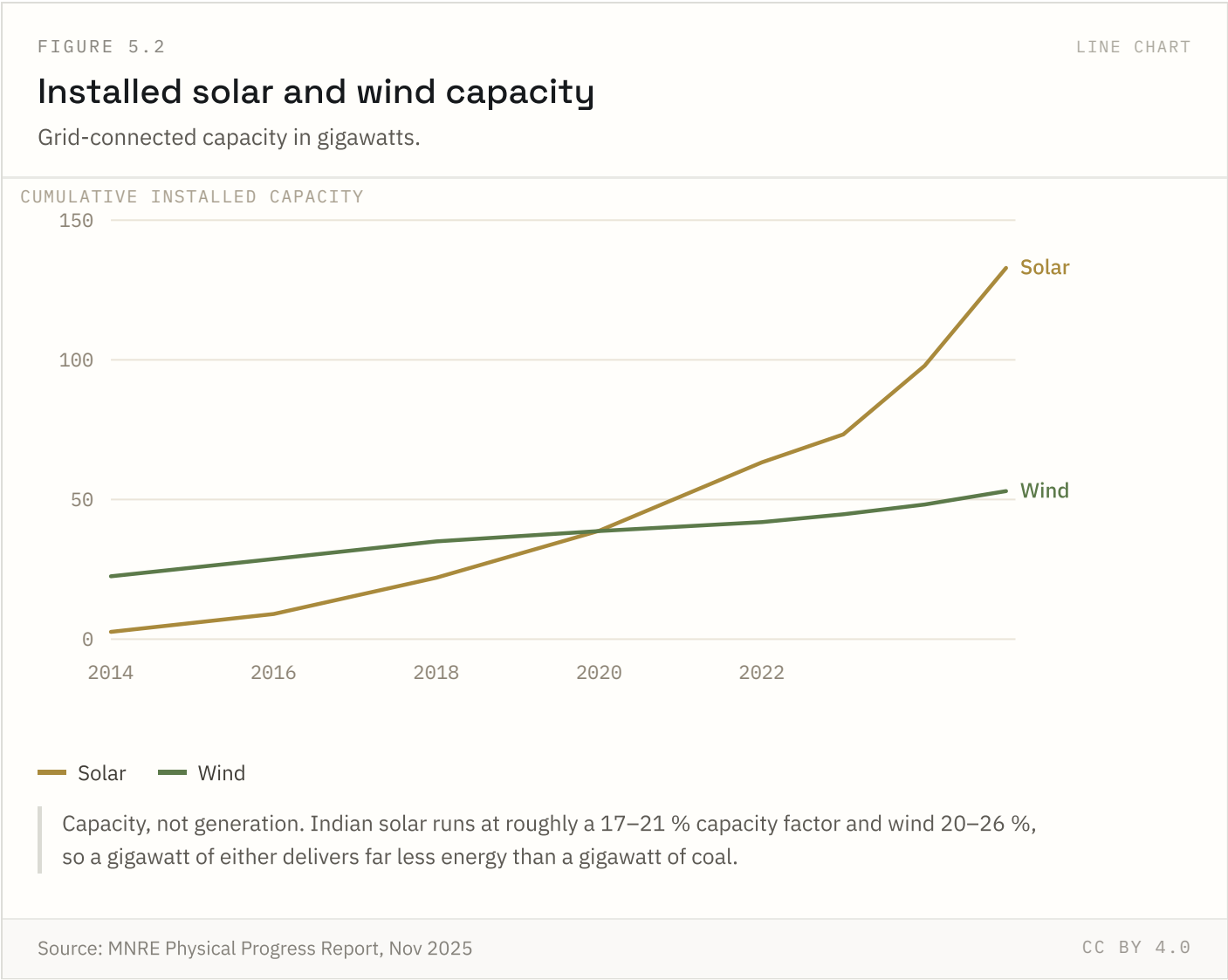
Energy & electricity

2 of 5 indicators in this chapter carry state-level detail; the rest are published at national level only.

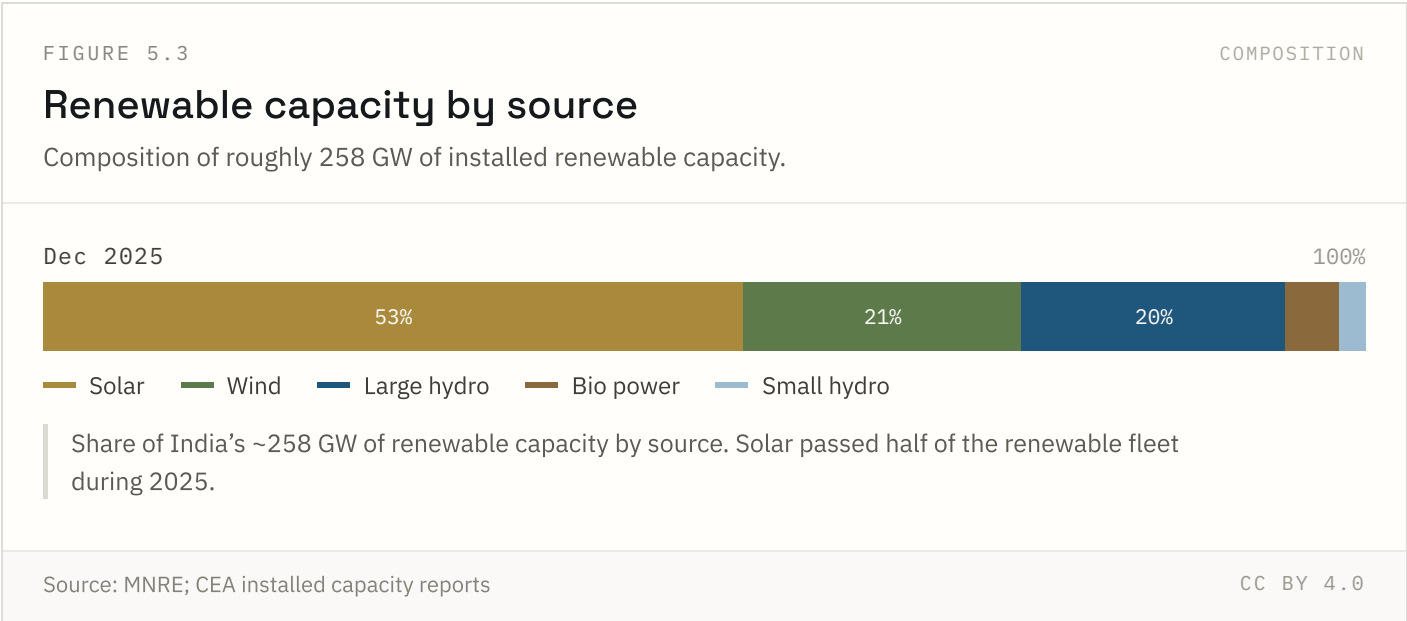
FIGURE 5.1 NATIONAL + STATE



Connection is not the same as supply. Administrative data reports near-universal connection since 2019, but survey data on hours of supply shows large rural shortfalls, especially in Bihar, Jharkhand and Uttar Pradesh.

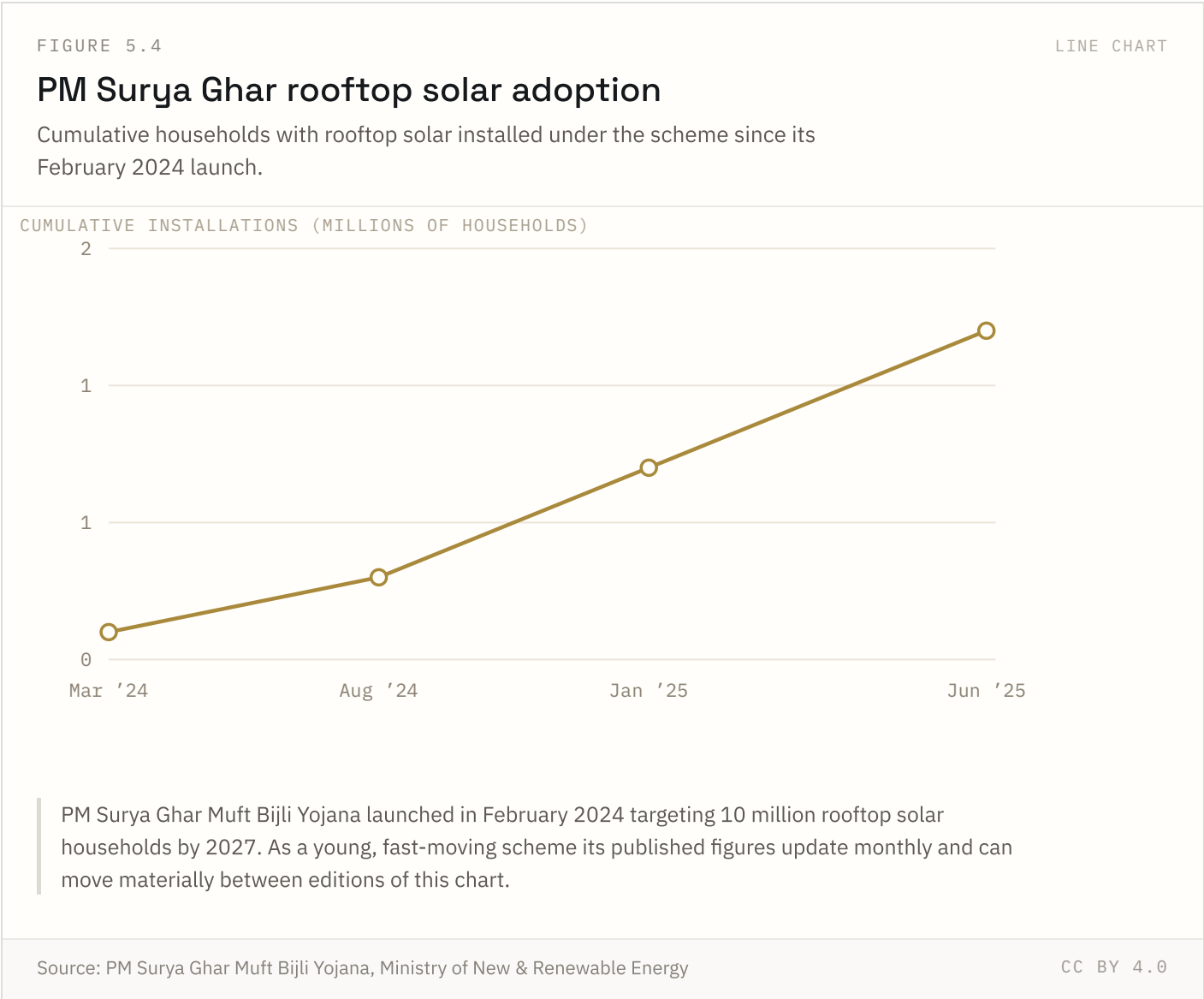


Capacity is not generation. Solar capacity factors in India run 17–21 per cent, wind 20–26 per cent, so a gigawatt of each delivers far less energy than a gigawatt of coal. Both series are ahead of the trajectory implied by the 2015 targets.



Solar passed half the renewable fleet during 2025. Large hydro, which took seven decades to build, is now smaller than solar, which took eleven years.

FIGURE 5.4 NATIONAL ONLY

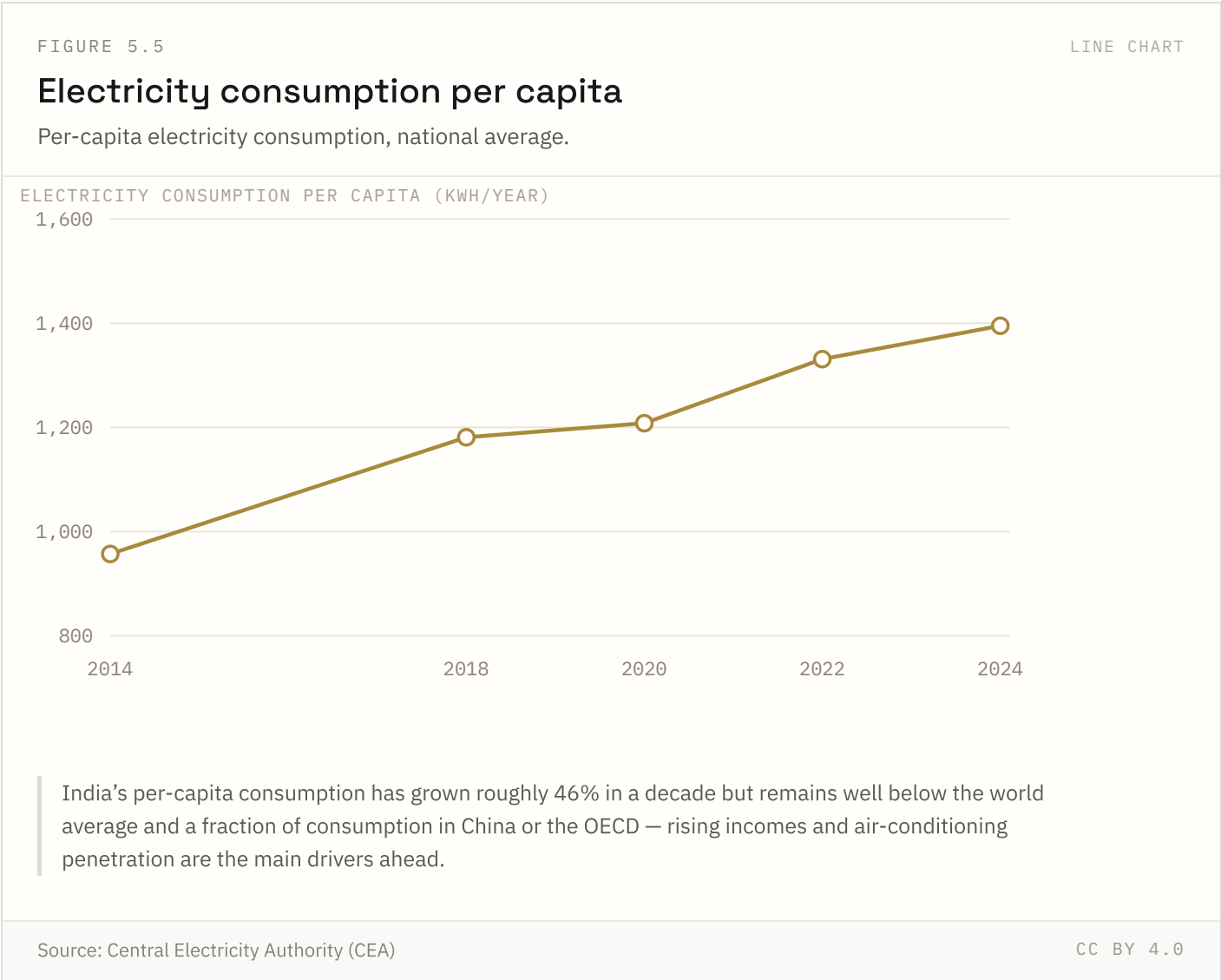


The scheme targets 10 million rooftop-solar households by 2027; adoption so far is a small fraction of that target and figures update frequently as installation data is reconciled with subsidy disbursement.

Source: PM Surya Ghar Muft Bijli Yojana, Ministry of New & Renewable Energy Coverage: Mar 2024–Jun 2025

Updated: 18 Aug 2026

FIGURE 5.5 NATIONAL ONLY



This is a different question from your household electricity access indicator: access asks whether a connection exists; this asks how much power the average person actually uses — a better proxy for living standards and industrial activity than connection rates alone.

Source: Central Electricity Authority (CEA) Coverage: 2014–2024 Updated: 18 Aug 2026

Methodology and limitations

Every figure in this report is reproduced from a published source without adjustment. India Insight Lab does not estimate, model or interpolate the values shown. Each chart carries the source it came from, the years it covers and the date the series was last updated.

Three limitations apply to everything above, and are stated rather than worked around:

- **Geographic depth.** Indicators reach national and, for 8 of 36 series, state level. District and city figures are not included because comparable data is not yet integrated.
- **Comparability across sources.** Series drawn from different sources use different definitions, reference periods and sampling designs. They are shown side by side but are not adjusted onto a common basis.
- **Currency.** Update dates vary by series; the most recent revision date is shown beneath each figure. Some official series lag by several years.

Nothing in this report is projected. Where a series ends, the report ends with it. No forecast, scenario or extrapolation has been added to extend a trend beyond its published data.

References

31 sources used in this report.

1	Atal Mission for Rejuvenation and Urban Transformation, MoHUA
2	CPCB National Air Quality Monitoring Programme
3	Census of India 2011
4	Census of India 2011, Primary Census Abstract for Slum
5	Census of India 2011; SRS 2021, RGI
6	Census of India, D-series migration tables
7	Census of India; Saubhagya dashboard, Ministry of Power
8	Census of India; UN World Population Prospects
9	Census of India; UN World Urbanization Prospects
10	Central Electricity Authority (CEA)
11	Central Ground Water Board (CGWB)
12	Global Carbon Budget
13	IQAir World Air Quality Report 2024
14	India Meteorological Department

15	India Meteorological Department, Statement on Climate of India
16	India State of Forest Report, Forest Survey of India
17	Indian Railways Yearbook, Ministry of Railways
18	MNRE Physical Progress Report, Nov 2025
19	MNRE; CEA installed capacity reports
20	NFHS-4 and NFHS-5, MoHFW
21	National Population Projections 2011–2036, Registrar General of India
22	PM Surya Ghar Muft Bijli Yojana, Ministry of New & Renewable Energy
23	Pradhan Mantri Awas Yojana dashboards, MoHUA & MoRD
24	SRS Abridged Life Tables 2017–21, RGI
25	SRS Statistical Report 2021, RGI
26	State drought declarations, compiled per the Manual for Drought Management
27	Swachh Bharat Mission-Urban dashboard, MoHUA
28	Swachh Survekshan, MoHUA
29	VIIRS-DNB satellite composites, research literature
30	Vahan Dashboard, Ministry of Road Transport & Highways
31	‘Road Accidents in India’, Ministry of Road Transport & Highways

Source URLs, licence terms and retrieval dates are not yet recorded for every series. Adding them is scheduled work, and until it is done these references name the publication rather than link to it.